International Power

Ernest Liu & David Y. Yang (draft of June 9, 2026)

A reader's guide and technical companion

Handout accompanying the presentation by Victor Xucheng Chen

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How to use this handout

- **Part I** (three pages) gives the whole paper at a glance: the two questions, the two answers, the headline numbers, and a map of how each piece of evidence supports the conclusions.
- **Parts II–IV** follow the paper's order — model, measurement, evidence — with every formula derived step by step.
- **Part V** is a technical toolkit: each method the paper uses (shift-share IV diagnostics, many-instrument estimators, shift-share standard errors, close-election IV, multiple-testing control, statistical multilateration, optimal transport, . . .) with motivation, formula, derivation and implementation in math and code. Labels T1–T28 match the backup slides.
- **Part VI** lists questions for discussion.

Basic undergraduate econometrics (OLS, standard errors, the idea of fixed effects and of 2SLS) is assumed. All numbers refer to the draft of June 9, 2026. Statements marked [ours] go beyond what the paper writes explicitly (our derivations or observations).

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Part I

The paper at a glance

1 The paper in five paragraphs

The question. International power is a country's capacity to shape outcomes by influencing other states' behaviour. Beyond military might, **commerce** can be a source of coercion: every sovereign controls its borders and can interrupt — or threaten to interrupt — its own exports and imports. Hirschman (1945) argued that trade creates relationships of dependence, influence and even domination, and named two sources of such power: *asymmetry* in dependence and the *cost of disruption*. History offers vivid cases: the 1973 Arab oil embargo against the US (the US imported about 35% of its oil from OPEC members; Arab producers controlled about 52% of world production) and China's 2010 halt of critical-mineral exports to Japan during a fishing-rights dispute (Japan sourced about 87% of its critical-mineral imports from China, which controlled about 97% of world production). Liu and Yang ask two questions: **(Q1)** does power that stems from trade change how countries engage and negotiate with each other, and **(Q2)** do countries strategically build such power when their relations deteriorate?

The model. In an illustrative model, import orders are placed *ex ante*. *Ex post*, a bilateral dispute may arise with probability λ_{ni}; a "coercer" i can then threaten to stop all its exports to a "target" n, which cannot immediately switch suppliers (a hold-up), and n can threaten to retaliate. Nash bargaining over the avoided trade war gives an expected transfer

$$\Delta_{ni} \;\approx\; \frac{1}{2}\sum_{k}\frac{1}{\theta_k}\left(s^k_{ni} - s^k_{in}\right),$$

the paper's definition of **international power** of i over n: the asymmetry in import dependence, weighted by the inverse of each sector's trade elasticity θ_k (hard-to-substitute goods count more). Two predictions follow: shocks that raise $|\Delta_{ni}|$ increase engagement and negotiation (P1), and shocks to geopolitical alignment prompt policies that build power (P2).

Measurement. The authors compute power for all country pairs from 2001 to 2021 using BACI trade data aggregated to 21 HS sections and Broda–Weinstein elasticities, and build two further datasets: bilateral **engagement** from 6,733,036 news-coded ICEWS events, and a high-frequency **geopolitical alignment** measure obtained by "GPS-style" multilateration of Gallup leadership-disapproval data combined with Polity scores. Power is related to, but clearly distinct from, differences in GDP, trade volumes and trade imbalances: it lives in the *sectoral composition* of dependence.

Findings. **(1)** Within country pairs over time, a one-standard-deviation rise in power asymmetry is followed by 0.236 more units of the engagement index (about +38.4% at the mean, +18.8% at the median); a shift-share instrument built from exporters' global market-share changes and importers' 1995–99 import mix gives 0.697. Countries also vote more alike in the UN General Assembly afterwards. Power from low-elasticity, complex and "critical" goods matters most; power from import dependence matters about twice as much as power from export dependence;

the effect vanishes among EU members. **(2)** When close-election turnover in country A shifts its alignment with country B, B raises its power over A: one standard deviation lower alignment raises B's power by 0.39 SD (from the median to about the 85th percentile), mostly by *cutting imports* from A through restrictive import policies.

Implications. Trade according to comparative advantage is positive-sum; accumulating power to extract geopolitical rents is negative-sum. Bilateral trade imbalances, often dismissed as accounting artefacts, can matter because they shape bargaining leverage — a precise version of the old mercantilist concern with "power and plenty" (Viner 1948).

Table 1: Headline numbers (draft of June 9, 2026).

Object	Estimate	s.e.	Source
Q1: power → engagement (all variables standardised within pair)			
Power_{t-1} on engagement_t, pair & year FE + controls	0.236^{***}	(0.0163)	Table 2, col. 3
Same, shift-share IV	0.697^{***}	(0.0589)	Table 2, col. 6
First-stage coefficient (F-statistic)	0.222^{***} (F = 3,996)	(0.00351)	Table A.4
Implied change in engagement	+38.4% (mean), +18.8% (median)		Sec. 4.3
UNGA vote similarity: OLS / IV	0.0105^{***} / 0.0350^{***}	(0.00263) / (0.0106)	Table A.10
Exporter vs. importer power (OLS col. 3)	0.234 vs. 0.134		Table A.15
Q2: alignment → power build-up (close-election IV)			
Turnover × pre-election alignment (first stage)	-0.137^{***}	(0.0251)	Table 3, Panel A
Predicted alignment on $\text{Power}_{B\to A}$ (2SLS)	-0.386^{***}	(0.147)	Table 3, Panel B
Magnitude	−1 SD alignment ⇒ +0.39 SD power		Sec. 5.3
B's import share from A / net liberal import policy	0.344^{**} / 0.356^{***}	(0.162) / (0.116)	Table 4
Measurement			
Corr. of baseline power with a full GE version	0.96		App. A.2
Corr. with GDP gap (no FE / pair & year FE)	.453 / .273		Table 1

2 The argument map

Figure 1 is the "mind map" of the paper: theory produces a measure; the measure is described and then used to answer one question about *consequences* and one about *causes*. Figures 2 and 3 show, step by step, how each conclusion is reached — and which alternative explanation each step rules out.

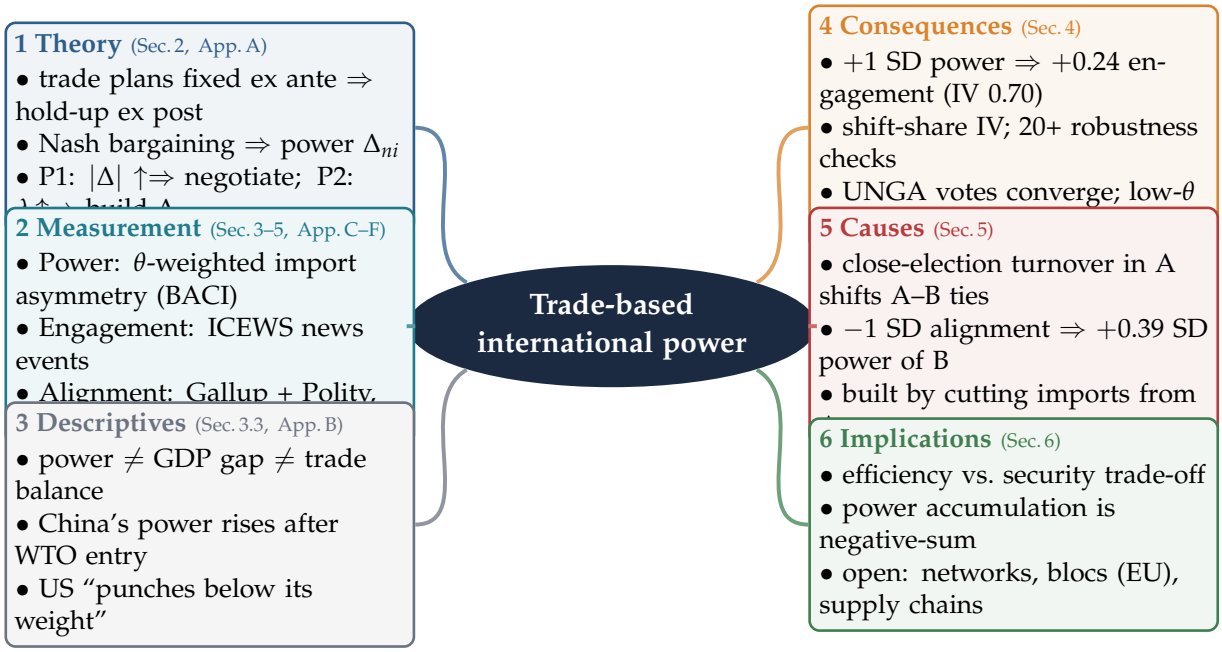


Figure 1: Paper map. Each box summarises one block of the paper; section numbers refer to the draft.

Reading the two paths. For Conclusion 1 the logical burden is to show that *power*, not size or trade volume or some common shock, drives engagement. Pair fixed effects remove everything time-invariant about a pair (geography, history, language); year effects remove global shocks; the GDP-gap and total-trade controls remove size and volume. The remaining worry is reverse causality or strategic, partner-specific trade (countries that plan to negotiate may also reshape their trade). The shift-share instrument answers it by using only power changes predicted by exporters' *global* market-share movements interacted with importers' *pre-1999* import mix. The diagnostics then ask whether the instrument leans on a few sectors (Rotemberg weights), whether those sectors' exposure correlates with other trends (balance tests), and whether inference is too optimistic (AKM). Mechanism tests check model-specific predictions: the effect should be larger for low-elasticity goods and for import (rather than export) dependence.

For Conclusion 2 the burden is to find shifts in alignment that do not themselves come from trade, and to separate one country's response from its partner's. Close elections in A provide shocks outside B's control; because a turnover moves A's alignment with different partners in opposite directions, A cannot easily tailor offsetting trade policies to each of them, while B can target A alone. Checks verify that turnovers are not predictable, have no *average* effect, and do not trigger targeted policies by A.

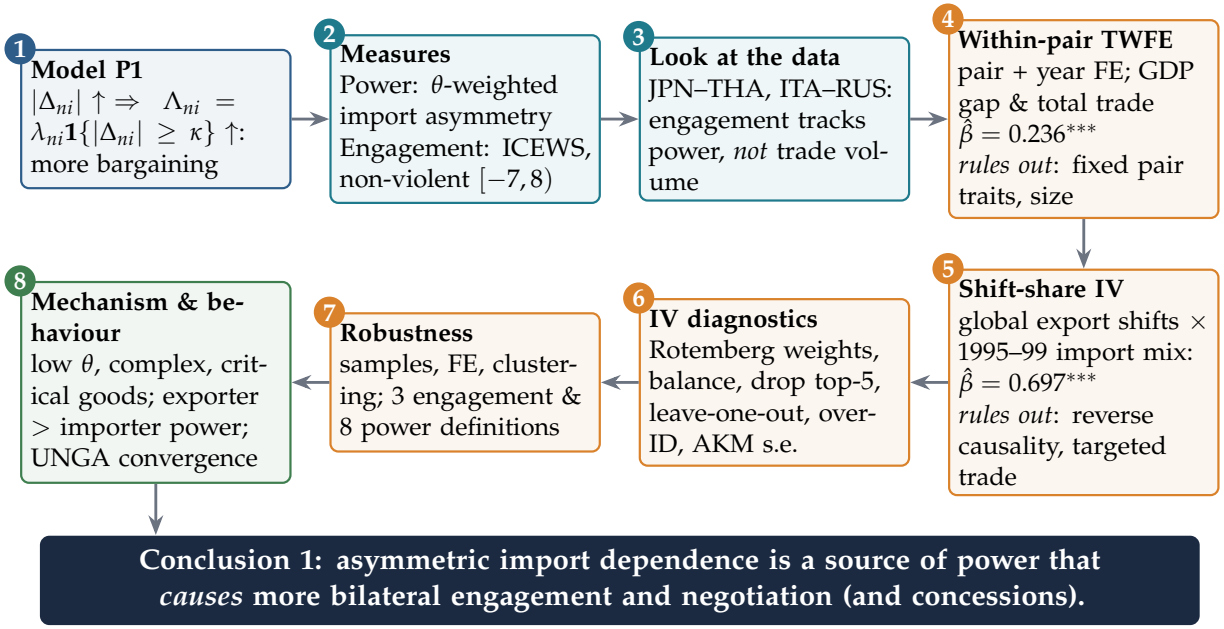


Figure 2: Path to Conclusion 1 (Section 4). Read left to right in the top row, then right to left in the bottom row.

3 Notation

Symbol	Meaning
n, i, j; k; t	countries (importer n, exporter i); sector; year
β_n^k	share of country n's tradable spending on sector k ($\sum_k \beta_n^k = 1$)
θ_k	trade elasticity of sector k; $\theta_k + 1$ = elasticity of substitution across origins
s_{ni}^k, π_{ni}^k	n's spending on i's good k (share of total tradable spending); within-sector share, $\pi_{ni}^k = s_{ni}^k / \beta_n^k$
δ_{ni}, $L_{ni} = -\delta_{ni}$	welfare change of n when i stops exporting to n; the corresponding loss
Δ_{ni}	model power of coercer i over target n (expected transfer from n to i)
λ_{ni}, κ, Λ_{ni}	dispute probability (inverse alignment); negotiation cost; negotiation probability
V_{nikt}	value of n's imports from i in sector k, year t (BACI)
$\bar{s}_{nit}$	n's elasticity-weighted import dependence on i (eq. 10)
$\text{Power}_{i\to n,t}$, $\text{Power}_{\{in\},t}$	directed power $\bar{s}_{nit} - \bar{s}_{int}$; undirected $\lvert\text{Power}_{i\to n,t}\rvert$
b_{nit}	shift-share instrument for $\bar{s}_{nit}$ (eq. 13)
Alignment_{ijt}	bilateral geopolitical alignment (Gallup + Polity)
$\text{TO}_{A,t}$, $\text{NT}_{A,t}$	close election in A in year t with / without turnover

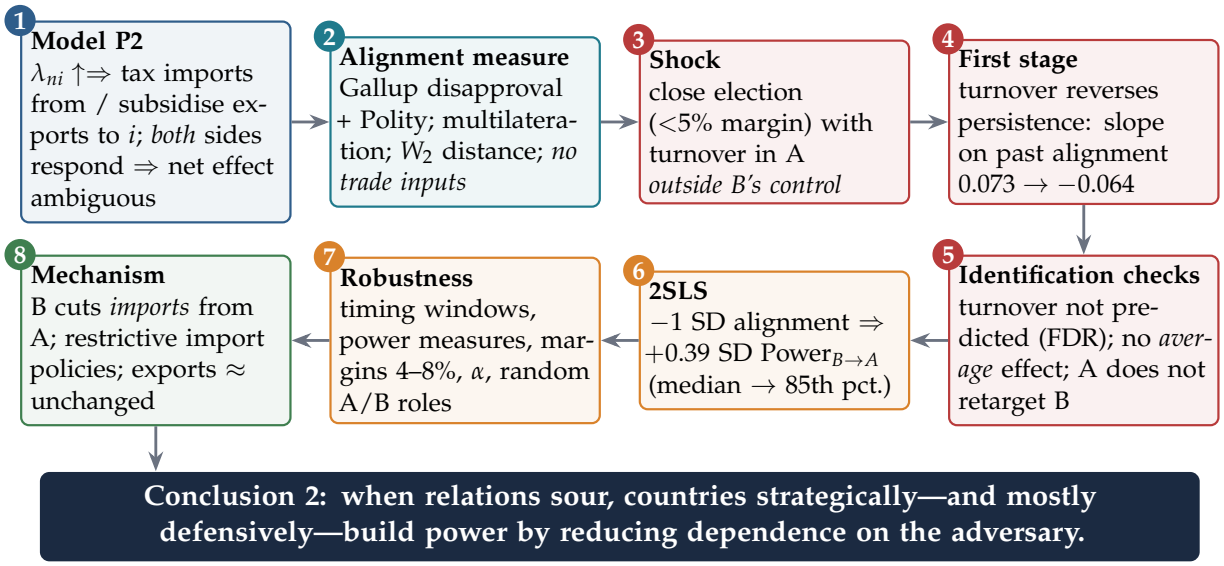


Figure 3: Path to Conclusion 2 (Section 5).

Part II

The model

4 Environment and equilibrium (T1)

There are N countries and K sectors. Each country has a representative consumer, competitive producers and a government that can negotiate with foreign governments.

Preferences. The consumer in n has quasi-linear utility over a freely traded outside good c_{n0} and differentiated goods, with Cobb–Douglas weights β_n^k across sectors and CES aggregation across origins within a sector:

$$u_n = \sum_k \beta_n^k \ln C_n^k + c_{n0}, \qquad C_n^k = \Big(\sum_i (c_{ni}^k)^{\rho_k}\Big)^{1/\rho_k}, \qquad \rho_k \equiv \frac{\theta_k}{\theta_k + 1}. \tag{1}$$

The elasticity of substitution across origins is $\sigma_k = 1/(1-\rho_k) = \theta_k + 1$; θ_k is the **trade elasticity**. The budget constraint is $p_{n0}c_{n0} + \sum_{i,k} p_{ni}^k c_{ni}^k = w_n \ell_n + T_n$, where T_n is a lump-sum transfer from the government.

Technology and prices. Country i produces its variety of good k with $q_i^k = a_i^k \ell_i^k$; shipping to n incurs iceberg costs $\tau_{ni}^k \geq 1$ ($\tau_{nn}^k = 1$). The outside good is produced one-for-one from labour in every country and traded freely, so wages equalise and are normalised to one: $w_n = p_{n0} = 1$. Competitive pricing gives $p_{ni}^k = \tau_{ni}^k / a_i^k$. The outside good settles any trade imbalance in differentiated goods.

Derivation of the expenditure shares (eq. 5)

With $w = p_0 = 1$ the consumer solves max $\sum_k \beta_n^k \ln C_n^k - \sum_{k,i} p_{ni}^k c_{ni}^k + \ell_n + T_n$.
Step 1 (within a sector). For a given level of C_n^k, cost minimisation over origins gives the CES demands

$$c_{ni}^k = \Big(\frac{p_{ni}^k}{P_n^k}\Big)^{-(\theta_k+1)} C_n^k, \qquad P_n^k = \Big(\sum_j (p_{nj}^k)^{-\theta_k}\Big)^{-1/\theta_k},$$

so that spending on the sector is $P_n^k C_n^k$.
Step 2 (across sectors). Choosing C_n^k to maximise $\beta_n^k \ln C_n^k - P_n^k C_n^k$ gives $P_n^k C_n^k = \beta_n^k$: sector spending is fixed, independently of income (quasi-linearity removes income effects), and total spending on differentiated goods is $\sum_k \beta_n^k = 1$.
Result. Spending on i's good k is

$$s_{ni}^k \equiv p_{ni}^k c_{ni}^k = \beta_n^k \Big(\frac{p_{ni}^k}{P_n^k}\Big)^{-\theta_k} = \beta_n^k \frac{(p_{ni}^k)^{-\theta_k}}{\sum_j (p_{nj}^k)^{-\theta_k}} \equiv \beta_n^k \, \pi_{ni}^k. \tag{5}$$

Because total tradable spending is one, s_{ni}^k is both a level and a share. The ratio $s_{ni}^k / s_{nj}^k = (p_{ni}^k / p_{nj}^k)^{-\theta_k}$ shows why θ_k is the trade elasticity: a 1% relative price increase lowers relative spending by θ_k%.

A useful identity (used in Section 5): since $c_{ni}^k \propto (p_{ni}^k)^{-(\theta_k+1)}$, we have $(c_{ni}^k)^{\rho_k} \propto (p_{ni}^k)^{-\theta_k}$, hence

$$\frac{(c_{ni}^k)^{\rho_k}}{\sum_j (c_{nj}^k)^{\rho_k}} = \pi_{ni}^k.$$

Timing and disputes. Import orders (prices and quantities) are placed ex ante. Before production, a bilateral dispute arises with probability λ_{ni} (undirected). The paper interprets λ_{ni} as (inverse) **geopolitical alignment**: low for allies, high for adversaries. In a dispute either country can pay a fixed cost κ to start negotiations; the initiator is the *coercer*, the other the *target*. The coercer threatens a comprehensive export stop; the target can threaten to retaliate. Disputes are off the equilibrium path, so prices and quantities are deterministic; only transfers depend on the threats (Figure 4).

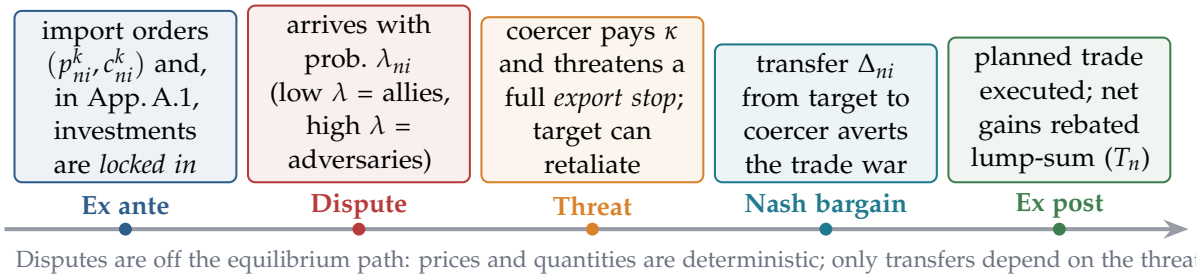


Figure 4: Timing of the model.

5 What an export stop costs the target (T2)

Suppose i stops all exports to n while every other order is delivered as planned. The change in n's welfare from sector k is (paper's eq. 6)

$$\delta_{ni}^k \equiv \beta_n^k \frac{\theta_k + 1}{\theta_k} \Big[\ln \sum_{j \neq i} (c_{nj}^k)^{\rho_k} - \ln \sum_j (c_{nj}^k)^{\rho_k} \Big] + p_{ni}^k c_{ni}^k \;\approx\; -\frac{1}{\theta_k} s_{ni}^k. \tag{6}$$

The first term is the lost utility from the CES bundle; the second is the money no longer spent on i's good, which goes to the outside good one-for-one.

Exact loss, first-order approximation and the timing assumption [ours]

Write $\pi \equiv \pi_{ni}^k$ and drop k. By the identity in Section 4, $\sum_{j \neq i} c_{nj}^{\rho} = (1-\pi) \sum_j c_{nj}^{\rho}$, so the bracket in (6) is $\ln(1-\pi)$, and $p_{ni} c_{ni} = \beta \pi$:

$$\frac{\delta^{\text{ante}}}{\beta} = \frac{\theta + 1}{\theta} \ln(1-\pi) + \pi.$$

Using $\ln(1-\pi) = -\pi - \pi^2/2 - \pi^3/3 - \cdots$,

$$\frac{\delta^{\text{ante}}}{\beta} = -\frac{\pi}{\theta} - \frac{\theta+1}{2\theta}\pi^2 + O(\pi^3) \quad \Longrightarrow \quad \delta^{\text{ante}} \approx -\frac{\beta\pi}{\theta} = -\frac{s_{ni}}{\theta}.$$

If instead the target could re-optimise after the cut-off, its price index would rise to $P' = \left(\sum_{j \neq i} p_{nj}^{-\theta}\right)^{-1/\theta} = P(1-\pi)^{-1/\theta}$; with indirect utility $\beta \ln(\beta/P) - \beta$,

$$\frac{\delta^{\text{post}}}{\beta} = \frac{1}{\theta} \ln(1-\pi) = -\frac{\pi}{\theta} + O(\pi^2).$$

This is the familiar "sufficient statistic" form of Arkolakis, Costinot and Rodríguez-Clare (2012), $\hat{W} = \hat{\lambda}^{1/\theta}$, applied to losing one supplier. The two timings agree to first order (an envelope-theorem result), and the gap is

$$\delta^{\text{post}} - \delta^{\text{ante}} = \beta\big[-\ln(1-\pi) - \pi\big] = \beta\Big[\frac{\pi^2}{2} + \frac{\pi^3}{3} + \cdots\Big] \geq 0,$$

second order — exactly the paper's "fifth interpretation". Summing over sectors (log utility makes losses additive across sectors): $\delta_{ni} = \sum_k \delta^k_{ni} \approx -\sum_k s^k_{ni}/\theta_k$.

Three messages follow. The loss is larger when n is more **dependent** on i (larger s^k_{ni}) and when the good is **harder to substitute** (smaller θ_k). And the **exporter** loses (almost) nothing: under competition its labour simply moves to the outside good (Section 9.1 shows the loss is second order even with sunk investments). This asymmetry — first-order losses for importers, second-order for exporters — is why power is defined through *import* dependence.

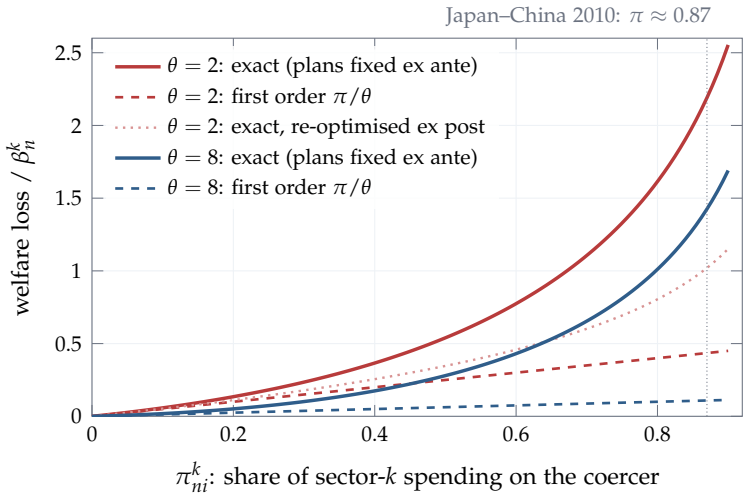


Figure 5: Welfare loss of losing one supplier, per unit of sector spending, as a function of the within-sector share π. Solid: exact loss with plans fixed ex ante; dashed: first-order approximation π/θ; dotted: exact loss with ex-post re-optimisation. The approximation is excellent for modest shares and understates losses at extreme dependence (self-made).

Watch out

The linear formula is designed for modest shares. With $\theta = 2$ and $\pi = 0.87$ (roughly Japan's dependence on Chinese critical minerals in 2010), the exact ex-ante loss is $\beta[1.5 \times 2.04 - 0.87] \approx 2.19\beta$, five times the first-order value 0.44β. At the HS-section level such extreme shares are rare, which the paper offers as one reason why the raw-unit relationship in Figure A.7 is mildly concave.

6 Bargaining and the definition of power (T3)

Conditional on a dispute, the two governments bargain over a transfer (or a policy concession) x from target n to coercer i that averts a mutual cut-off. Write the losses as positive numbers, $L_{ni} \equiv -\delta_{ni}$ and $L_{in} \equiv -\delta_{in}$.

Nash bargaining solution

Payoffs relative to the status quo: agreement gives n the payoff $-x$ and i the payoff $+x$; disagreement (the trade war) gives $-L_{ni}$ and $-L_{in}$. The generalised Nash solution with bargaining weight b for n solves

$$\max_x \, (L_{ni} - x)^{\,b} \, (L_{in} + x)^{1-b}.$$

The first-order condition $(1-b)(L_{ni} - x) = b(L_{in} + x)$ gives $x^* = (1-b)L_{ni} - bL_{in}$. With symmetric weights $b = \frac{1}{2}$,

$$\Delta_{ni} \equiv x^* = \tfrac{1}{2}\big(L_{ni} - L_{in}\big) \;\approx\; \frac{1}{2}\sum_k \frac{1}{\theta_k}\Big(s^k_{ni} - s^k_{in}\Big). \tag{7}$$

Each side gains $\frac{1}{2}(L_{ni} + L_{in})$ relative to the trade war: the avoided deadweight loss is split equally, and the transfer reflects only the *difference* in what each side would lose.

Properties. (i) *Antisymmetry*: $\Delta_{ni} + \Delta_{in} = 0$, so power is zero-sum within a pair. (ii) *Ex ante*: Δ is computed from planned flows and is the expected transfer the coercer can extract *should* a dispute arise — the threat need not be executed. (iii) It combines Hirschman's two sources of power in one number: *asymmetry* $(s^k_{ni} - s^k_{in})$ and *cost of disruption* $(1/\theta_k)$. (iv) The bargain is reduced-form: it holds fixed the rest of the world, including simultaneous negotiations with third countries; a symmetric Nash split can be micro-founded by alternating offers with equally patient players (Rubinstein 1982).

Sign convention

Equation (6) defines δ as a welfare *change* (negative), while equation (7) writes $\Delta_{ni} = \frac{1}{2}(\delta_{ni} - \delta_{in})$ with δ read as the size of the loss. Writing $L = -\delta$, as here, removes the ambiguity: the target pays when its loss exceeds the coercer's.

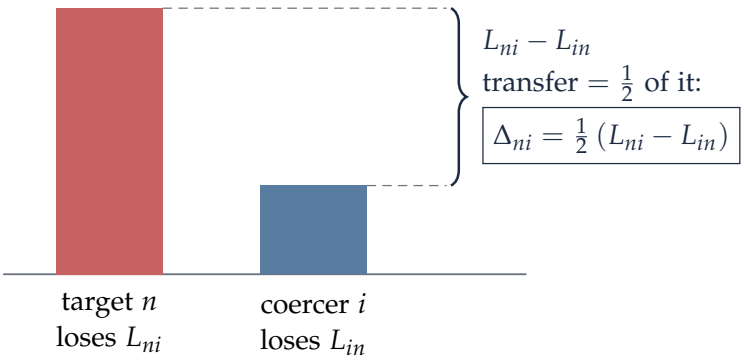


Figure 6: The target pays half the difference between the two losses (self-made). Disagreement point = mutual trade war.

7 Negotiation costs, expected payoffs and ex-ante policy (T4)

Who negotiates. When a dispute arises, the country that gains from bargaining pays κ to start it if its expected gain covers the cost: i initiates against n iff $\Delta_{ni} \geq \kappa$. Because Δ is antisymmetric, at most one side initiates, and the probability of a negotiation is

$$\Lambda_{ni} \equiv \lambda_{ni} \cdot \mathbf{1}\{|\Delta_{ni}| \geq \kappa\}. \tag{8a}$$

A smooth version[ours]: if the cost κ varies across episodes with distribution G, the probability becomes $\lambda_{ni}G(|\Delta_{ni}|)$, which rises smoothly with $|\Delta_{ni}|$ — a convenient rationale for regressing engagement on |Power| linearly.

Expected payoff. To first order, the expected payoff of n from disputes is

$$L_n \equiv -\sum_i \Lambda_{ni}\Big(\Delta_{ni} + \kappa\,\mathbf{1}\{\Delta_{in} \geq \kappa\}\Big)\,: \tag{8}$$

n pays Δ_{ni} to stronger partners (receives when $\Delta_{ni} < 0$) and pays κ when it initiates.

Ex-ante welfare and the optimal policy wedges (eq. 9)

Insert (7) into (8), using $s^k_{ni} = p^k_{ni}c^k_{ni}$ (spending on imports from i) and $s^k_{in} = p^k_{in}c^k_{in} = p^k_n\tau^k_{in}c^k_{in} = p^k_n q^k_{in}$ (the value of n's exports to i at producer prices):

$$L_n = -\sum_{k,i}\frac{\Lambda_{ni}}{2\theta_k}\,p^k_{ni}c^k_{ni} + \sum_{k,i}\frac{\Lambda_{ni}}{2\theta_k}\,p^k_n q^k_{in} - \sum_i \Lambda_{ni}\kappa\,\mathbf{1}\{\Delta_{in} \geq \kappa\}.$$

Replacing T_n by L_n in utility gives the government's ex-ante welfare

$$W_n = \sum_k \beta^k_n \ln C^k_n - \sum_{k,i}\Big(1 + \frac{\Lambda_{ni}}{2\theta_k}\Big)p^k_{ni}c^k_{ni} + \sum_{k,i}\frac{\Lambda_{ni}}{2\theta_k}p^k_n q^k_{in} + \ell_n - \sum_i \Lambda_{ni}\kappa\,\mathbf{1}\{\Delta_{in} \geq \kappa\}. \tag{9}$$

Imports. A price-taking planner (holding Λ locally fixed) sets the marginal utility of c^k_{ni} equal to $(1 + \Lambda_{ni}/2\theta_k)\,p^k_{ni}$, whereas a private consumer sets it equal to the price paid. An ad-valorem import tariff η^k_{ni} (revenue rebated) makes the consumer pay $(1+\eta^k_{ni})p^k_{ni}$, so the planner's allocation is implemented by

$$\eta^k_{ni} = \frac{\Lambda_{ni}}{2\theta_k}.$$

Exports. An extra unit of q^k_{in} is worth p^k_n to private exporters but $p^k_n(1 + \Lambda_{ni}/2\theta_k)$ to society, because it raises n's power over i; an export subsidy $t^k_{ni} = \Lambda_{ni}/(2\theta_k)$ closes the gap.

The wedges are larger against **adversaries** (high λ_{ni}) and in **hard-to-substitute** sectors (low θ_k), and vanish when $|\Delta_{ni}| < \kappa$. Decentralised agents ignore power, so they *over-import* from and *under-export* to adversaries. With irreversible sector-specific investments (Appendix A.1), the same logic applies to industrial policy targeting those investments.

Watch out

Three caveats on the policy result: Λ jumps at $|\Delta| = \kappa$, so the FOCs hold only away from the threshold; the policy changes Δ itself (the planner is "price-taking" in Λ); and the partner retaliates with mirror-image policies. The last point is why the effect of an alignment shock on *net* bilateral power is ambiguous.

8 Two predictions and the identification logic

The two testable predictions

P1 (consequences). Shocks to preferences or technology (β_n^k, a_i^k) that raise power asymmetry $|\Delta_{ni}|$ raise the probability $\Lambda_{ni} = \lambda_{ni}\mathbf{1}\{|\Delta_{ni}| \geq \kappa\}$ of engagement and negotiation.
P2 (causes). Shocks to bilateral relations (λ_{ni}) prompt policies that build bilateral power.

Under P2 both countries respond, in opposite directions, so the effect on the net quantity Δ_{ni} is ambiguous. It becomes unambiguous if one isolates *one* country's response while holding the partner's fixed: a more adversarial relationship then prompts that country to raise its power over the partner. Section 16 implements this by studying how B's power over A responds to alignment shifts generated by A's domestic elections.

Table 2: From model objects to empirical counterparts.

Model	Empirical counterpart
Δ_{ni} (power of i over n)	$\text{Power}_{i\to n,t} = \bar{s}_{nit} - \bar{s}_{int}$; undirected |Power| when the outcome is undirected
Λ_{ni} (negotiation probability)	ICEWS engagement index (sum of within-pair z-scores of event counts)
λ_{ni} (dispute probability)	minus geopolitical alignment (Gallup + Polity)
shocks to β_n^k, a_i^k	shift-share instrument: exporters' global market shares × importers' 1995–99 import mix
shocks to λ_{ni}	close-election turnover in partner A
one-sided response	B's directed power over A, B's import/export shares and targeted policies

9 How robust is the measure? (Appendix A and Section 2 remarks)

9.1 Irreversible investment: exporter losses are second order (T5)

In the baseline, a cut-off costs the exporter nothing because labour is reallocated ex post. Appendix A.1 adds an irreversible input committed before disputes realise. The firm in i produces good k with

$$q = a\left(\frac{\iota}{1-\alpha}\right)^{1-\alpha}\left(\frac{x}{\alpha}\right)^{\alpha}, \tag{22}$$

where ι is the ex-ante investment and x the flexible input, both made from labour. With competitive pricing $p = 1/a$, the ex-ante choice is $\iota = (1-\alpha)pq$ (cost share).

Producer loss after losing a fraction γ of sales[ours]

Given ι, producing $\tilde{q}$ requires $\tilde{x} = \alpha(\tilde{q}/a)^{1/\alpha}\big(\iota/(1-\alpha)\big)^{-(1-\alpha)/\alpha}$. Since $\iota/(1-\alpha) = pq = q/a$,

$$\tilde{x} = \alpha\,\frac{q}{a}\left(\frac{\tilde{q}}{q}\right)^{1/\alpha}, \qquad x = \alpha\,\frac{q}{a} = \alpha pq.$$

Quasi-rents are $p\tilde{q} - \tilde{x}$ with the sanction and $pq - x = (1-\alpha)pq$ without. If the importer was to buy a fraction γ of output, $\tilde{q} = (1-\gamma)q$ and the loss is

$$\text{Loss} = pq\big[\gamma - \alpha + \alpha(1-\gamma)^{1/\alpha}\big].$$

A second-order expansion, $(1-\gamma)^{1/\alpha} = 1 - \frac{\gamma}{\alpha} + \frac{1}{\alpha}\left(\frac{1}{\alpha}-1\right)\frac{\gamma^2}{2} + O(\gamma^3)$, gives

$$\text{Loss} = pq\,\frac{1-\alpha}{2\alpha}\,\gamma^2 + O(\gamma^3).$$

> The first-order terms cancel (the paper shows exactly this cancellation); the explicit second-order coefficient $\frac{1-\alpha}{2\alpha}$ is our addition.

The intuition is the envelope theorem: competitive firms earn no pure rents and the flexible input re-optimises, so a small loss of market size has only a second-order effect on profits. The importer, by contrast, loses to *first* order in its expenditure share. The conclusion can change with market power and markups (losing a market then costs first-order profits) or with endogenous factor prices; that is why the paper also tests "importer power" built from export dependence (Section 15.6.1).

9.2 General equilibrium: a multi-sector Ricardian model (T6)

Appendix A.2 computes the general-equilibrium counterpart of power in the multi-sector Ricardian model of Costinot, Donaldson and Komunjer (CDK), using "exact hat algebra" (Dekle, Eaton and Kortum 2008).

Model. Fréchet productivity with shape θ_k gives expenditure shares and trade balance

$$S_{ni}^k = \frac{(w_i \tau_{ni}^k / z_i^k)^{-\theta_k}}{\sum_j (w_j \tau_{nj}^k / z_j^k)^{-\theta_k}}, \qquad w_i \ell_i = \sum_n \sum_k \beta_n^k w_n \ell_n S_{ni}^k, \tag{26–27}$$

and price indices $P_n^k = [\sum_i (w_i \tau_{ni}^k / z_i^k)^{-\theta_k}]^{-1/\theta_k}$, $P_n = \prod_k (P_n^k)^{\beta_n^k}$.

Hat algebra. Writing $\hat{x} = x'/x$ and holding productivities fixed, the counterfactual equilibrium only requires baseline shares, spending weights, incomes and θ:

$$\hat{S}_{ni}^k = \frac{(\hat{w}_i \hat{\tau}_{ni}^k)^{-\theta_k}}{\sum_j S_{nj}^k (\hat{w}_j \hat{\tau}_{nj}^k)^{-\theta_k}}, \qquad \hat{w}_i w_i \ell_i = \sum_{n,k} \beta_n^k \hat{w}_n w_n \ell_n S_{ni}^k \hat{S}_{ni}^k, \tag{30–31}$$

$$\hat{P}_n^k = \Big[\sum_i S_{ni}^k (\hat{w}_i \hat{\tau}_{ni}^k)^{-\theta_k} \Big]^{-1/\theta_k}, \qquad \hat{W}_n = \frac{\hat{w}_n}{\prod_k (\hat{P}_n^k)^{\beta_n^k}}. \tag{32–33}$$

Why it works: the unknown levels of z_i^k and τ_{ni}^k enter the baseline shares; dividing the counterfactual by the baseline cancels them.

The bilateral cut-off. Set $(\hat{\tau}_{ab}^k)^{-\theta_k} = (\hat{\tau}_{ba}^k)^{-\theta_k} = 0$ for all k (all other $\hat{\tau} = 1$), solve for $\hat{w}$, and compute real-income losses $\Delta_a = 1 - \hat{W}_a$, $\Delta_b = 1 - \hat{W}_b$. The GE analogue of power is the differential loss $\Delta_a - \Delta_b = \hat{W}_b - \hat{W}_a$ (eq. 37). Across pairs, it has a **correlation of 0.96** with the baseline measure: terms-of-trade effects are small relative to the first-order partial-equilibrium channel.

Implementation: solving the cut-off counterfactual (Python sketch)

```
import numpy as np

def cutoff_losses(S, beta, Y, theta, a, b, tol=1e-12, damp=0.5, maxit=10000):
    """S[n,i,k]: baseline expenditure shares (sum over i = 1); beta[n,k]: sector
    weights; Y[n] = w_n * l_n; theta[k]; a, b: the two countries that stop trading.
    Returns the real-income loss 1 - W_hat for every country."""
    N, _, K = S.shape
    tmt = np.ones((N, N, K))                # (tau_hat)^(-theta); zero = no trade
    tmt[a, b, :] = 0.0; tmt[b, a, :] = 0.0
    w = np.ones(N)
```

```
    for _ in range(maxit):
        num = S * (w[None, :, None] ** (-theta[None, None, :])) * tmt
        den = num.sum(axis=1, keepdims=True)
        S_new = num / den                    # counterfactual shares S * S_hat
        income = np.einsum('nk,n,nik->i', beta, w * Y, S_new)
        w_new = income / Y
        w_new = w_new / w_new[0]             # numeraire
        if np.max(np.abs(w_new - w)) < tol:
            w = w_new; break
        w = damp * w_new + (1 - damp) * w    # damping helps convergence
    P_hat = np.exp((beta * np.log(den[:, 0, :] ** (-1.0 / theta[None, :]))).sum(axis=1))
    return 1.0 - w / P_hat                   # Delta_c; GE power analogue = Delta_a -
    Delta_b
```

Real income is invariant to the choice of numeraire (scaling all $\hat{w}$ scales all $\hat{P}$). For all pairs, loop over (a, b) — about $200^2/2$ counterfactuals, each converging in a few hundred iterations.

9.3 Targeted sanctions and type-I extreme-value shocks (T7)

The baseline lets the coercer threaten all sectors. Suppose instead it can cut only *one* sector, choosing the most damaging one subject to sector-specific political shocks $\sigma\varepsilon_k$ with ε_k i.i.d. Gumbel. The paper states that the expected maximum damage yields a measure "isomorphic to (7) up to a scaling constant". One way to see it[ours]: with $d_k = s_{ni}^k/\theta_k$,

$$\mathbb{E}\Big[\max_k\{d_k + \sigma\varepsilon_k\}\Big] = \sigma\gamma_E + \sigma\ln\sum_k e^{d_k/\sigma} \;\approx\; \sigma(\gamma_E + \ln K) + \frac{1}{K}\sum_k \frac{s_{ni}^k}{\theta_k} \qquad (d_k \ll \sigma),$$

using the log-sum-exp formula for Gumbel maxima and $\ln\sum_k e^{x_k} \approx \ln K + \frac{1}{K}\sum_k x_k$ for small x_k ($\gamma_E \approx 0.5772$). The same constant appears in the retaliation term, so it cancels in the Nash difference: $\Delta_{ni}^{\text{targeted}} \approx \Delta_{ni}/K$. When political shocks are small relative to damages, the maximum dominates instead; the empirical "max-sector" measures capture that case.

9.4 Size, timing and the direction of threats

- **Shares vs. levels.** In the model total tradable spending is one everywhere. If the cost of concessions scales with economic size (giving up intellectual property is costlier for a large economy), share-based measures are appropriate; if concessions are fixed payments, *level*-based measures are. The paper implements both.
- **Ex-ante vs. ex-post sourcing.** The two timings agree to first order (Section 5). The ex-ante formulation is kept because it makes governments' incentives to shape *ex-ante exposure* transparent.
- **Export stops vs. import bans.** Under competition, an import ban raises the banning country's price index (a first-order loss for the banner) while leaving the exporter's welfare unchanged, so export-withholding threats dominate. With markups this can change; hence the "importer power" robustness test.

Part III

Measurement and data

10 The power measure

10.1 From Δ_{ni} to data

Following equation (7), importer n's elasticity-weighted dependence on exporter i in year t is

$$\bar{s}_{nit} = \sum_{k \in \mathcal{K}} \frac{1}{\theta_k} \frac{V_{nikt}}{\sum_{i'} \sum_{k'} V_{ni'k't}}, \tag{10}$$

and the (directed) **power** of i over n is

$$\text{Power}_{i \to n,t} = \bar{s}_{nit} - \bar{s}_{int}. \tag{11}$$

It is antisymmetric, so its mean across directed pairs is zero by construction; for undirected outcomes the paper uses $\text{Power}_{\{in\},t} = |\text{Power}_{i \to n,t}|$. Among the 13,244 pairs in the sample, 6,670 see the sign of directed power flip at least once.

Data choices.

- **Trade flows** V_{nikt}: BACI (CEPII), which reconciles importer and exporter reports in UN Comtrade at the HS 6-digit level for about 200 countries, 1995–2021. The analysis uses **2001–2021**: in 2000 BACI lowered the threshold above which it records a positive flow, which creates artificial jumps (including 1995–2000 shrinks the estimates, Table A.9 Panel B.2).
- **Sectors** $\mathcal{K}$: the 21 HS **sections**, with two adjustments: chapters 25 (salt, sulphur, lime, cement) and 26 (ores, slag, ash) are moved from "minerals" (mostly petroleum) to "base metals"; arms and ammunition are taken from SIPRI, whose coverage is much better than customs data.
- **Elasticities** θ_k: Broda and Weinstein (2006) estimates at the HTS level, averaged within HS 6-digit codes and then aggregated to sections with 2001–2021 world trade weights (Section 11).
- **Denominator**: n's *total imports*, not its total expenditure as in the model — sector-level domestic expenditure is rarely observed. An Eora-based variant adds imputed domestic spending (Section 10.3).

Implementation: computing power from a trade panel (Python/pandas)

```
import pandas as pd
# V: one row per (imp, exp, sec, year) with trade value 'value' (BACI aggregated to HS
    sections)
# theta: pandas Series indexed by 'sec' (sector trade elasticities)
V = V.merge(theta.rename('theta'), left_on='sec', right_index=True)
tot_imp = V.groupby(['imp', 'year'])['value'].transform('sum')     # n's total imports in t
V['dep'] = V['value'] / V['theta'] / tot_imp                        # (1/theta_k) V_nikt /
```

```
    sum V_n..t
sbar = V.groupby(['imp', 'exp', 'year'])['dep'].sum()              # sbar_{nit}, eq. (10)
# reverse direction: value of sbar_{int} placed on row (n, i, t)
rev = sbar.copy()
rev.index = rev.index.set_names(['exp', 'imp', 'year'])
rev = rev.reorder_levels(['imp', 'exp', 'year'])
power = sbar.sub(rev, fill_value=0)                                # Power_{i -> n, t},
    eq. (11)
abs_power = power.abs()                                            # undirected version
```

10.2 Eight alternative measures

Table 3 lists the variants used throughout the robustness analysis, with their correlation with the baseline measure (Table 1, Panel A). Except for the *level* version — which embodies a different assumption about how concessions scale with size — all are highly correlated with the baseline.

Table 3: Alternative power measures and their correlation with the baseline.

Variant	Construction	no FE	pair & year FE
Equally weighted sectors	drop $1/\theta_k$ (all sectors weigh the same)	.978	.924
Max-disruption sector	only the sector maximising V_{nikt} (an embargo on one sector)	.863	.805
Max sector, θ-weighted	only the sector maximising V_{nikt}/θ_k	.761	.677
HS 2-digit	sectors = HS chapters, elasticities aggregated to chapters	.983	.925
Levels	$\log(1+\bar{s}^{\text{level}}_{nit}) - \log(1+\bar{s}^{\text{level}}_{int})$ with elasticity-weighted dollar values	.204	.117
Actually traded sectors	$\mathcal{K}$ = sectors traded between the pair	.890	.569
Domestic expenditure	add imputed domestic spending to the denominator (Eora)	.806	.643
Trade in services	add an imputed services "sector" (Eora)	.924	.846

Summary statistics. Over all directed pairs the baseline has mean 0 and standard deviation 0.001; conditional on being positive, mean 0.003 and SD 0.003. Power is a small number because it is a difference of two shares of total imports, each divided by θ_k (typically between about 3 and 30).

10.3 Eora imputations (T27)

The Eora26 multi-region input–output database (184 countries, 26 sectors; freely available up to 2017, so the 2015–17 average is used) records intranational transactions and services.

Domestic expenditure (Appendix C.1). For importer n and HS 6-digit product g in Eora sector $k = k(g)$, domestic purchases are imputed by scaling observed imports with Eora's domestic-to-foreign sourcing ratio,

$$V_{nng} = \Big(\sum_{i \in \text{Foreign}} V^{\text{BACI}}_{nig}\Big) \times \frac{V^{\text{Eora}}_{nnk}}{\sum_{i \in \text{Foreign}} V^{\text{Eora}}_{nik}}, \tag{40}$$

and then rescaled so that national totals match Eora, $V^{\text{norm}}_{nng} = V_{nng}\, V_{nn} \big/ \sum_g V_{nng}$ (eqs. 41–42). Two problems are addressed: mismatches between Eora and BACI trade volumes (Liberia,

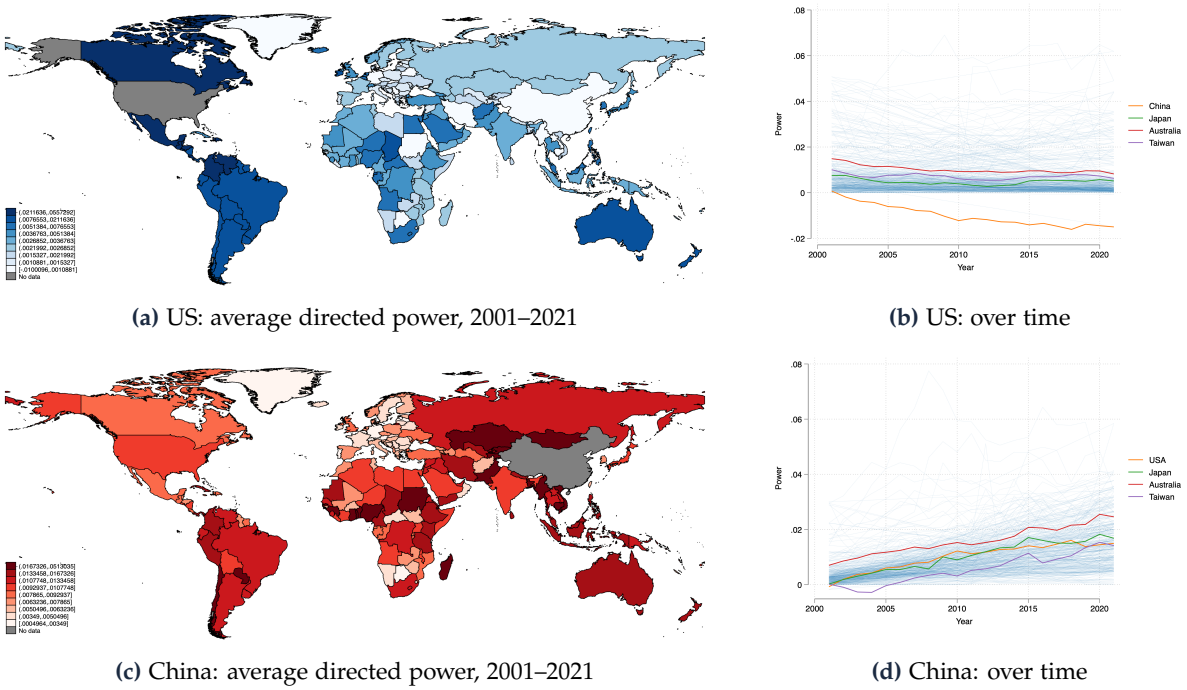
Madagascar, Myanmar and Somalia, whose Eora/BACI ratio is extreme in at least 20 of 21 years, are corrected with yearly median ratios) and the coarse-to-fine mapping from 26 sectors to HS-6 products (handled by the national rescaling).

Services (Appendix C.2). Each Eora sector is classified as "non-goods" if BACI covers less than 10% of its Eora trade (construction, education/health, finance and business services, hotels and restaurants, maintenance, post and telecom, private households, public administration, recycling, retail, transport, wholesale). For each pair the non-goods-to-goods ratio

$$\gamma_{ni} = \frac{\sum_{k\in\text{non-goods}} V_{nik}^{\text{Eora}}}{\sum_{k\in\text{goods}} V_{nik}^{\text{Eora}}}, \qquad V_{ni,\text{services}} = \gamma_{ni} \sum_{k\in\text{BACI}} V_{nik}, \tag{43–44}$$

creates one extra sector, assigned the average elasticity of all other sectors.

10.4 Descriptive patterns



(a) US: average directed power, 2001–2021 **(b)** US: over time

(c) China: average directed power, 2001–2021 **(d)** China: over time

Figure 7: Who holds power over whom (paper's Figure 1).

- **The US** holds disproportionate power over the Americas and over allies such as Japan and Australia, but little over China despite the enormous bilateral trade; its power over China declined, especially in 2000–2005, and turned negative (Figure 7).
- **China**'s power rose against most countries after WTO accession: neighbours (India, Vietnam), East Africa, South America. Countries more dependent on China tend to depend less on the US: regressing China's power over i on the US's power over i gives about -0.14 ($p < 0.001$).
- **Power vs. GDP** (Figure A.3): China wields more power than its GDP gap would predict — even positive power over the US (2001–2020) and Japan (2001–2010) while its GDP was smaller. The US "punches below its economic weight" in Asia, Europe, Africa and Oceania (although above the global fit in the Americas).

- **Power vs. trade balance** (Figure A.2): most coercer–target-years have a balance near zero, yet power varies widely across partners and over time.
- **Power vs. other sources of influence** (Table 1, Panels B–C): positive but weak correlations with gaps in GDP (.453; .273 with pair and year effects), GDP per capita (.135; −.013), military spending (.36; .192), aid dependence (.094; .038) and sovereign-debt exposure (.335; .193). The correlation with the FBIC "political bandwidth" index falls from .29 to .02 with fixed effects, because FBIC moves slowly while trade power changes year to year.
- **Blocs** (Figure A.19): each pre-2000 EU member has *negative* power over China, but the EU treated as one entity (excluding intra-EU trade) has *positive* power over China — a reminder that the bilateral measure ignores coalitions.

11 Trade elasticities (T8)

The weights $1/\theta_k$ come from estimated elasticities of substitution. The baseline uses **Broda and Weinstein (2006)**, who apply the method of **Feenstra (1994)** to US imports of each HTS good g from supplier countries c.

The Feenstra / Broda–Weinstein estimator in brief

Demand for each variety is CES with elasticity σ_g; export supply is upward-sloping with inverse elasticity ω_g. Taking first differences over time and then differences relative to a reference supplier r (Δ^r) removes the unobserved price index and common shocks. Multiplying the differenced demand and supply equations, under the assumption that demand and supply shocks are uncorrelated, gives an estimating equation for each good:

$$\underbrace{(\Delta^r \ln p_{gct})^2}_{Y_{gct}} = \vartheta_1 \underbrace{(\Delta^r \ln s_{gct})^2}_{X_{1,gct}} + \vartheta_2 \underbrace{(\Delta^r \ln p_{gct})(\Delta^r \ln s_{gct})}_{X_{2,gct}} + u_{gct},$$

$$\vartheta_1 = \frac{\omega_g}{(1+\omega_g)(\sigma_g - 1)}, \qquad \vartheta_2 = \frac{1 - \omega_g(\sigma_g - 2)}{(1+\omega_g)(\sigma_g - 1)},$$

where p are unit values and s import shares. Identification comes from *heteroskedasticity* across suppliers: countries with more volatile supply shocks trace out the demand curve. (σ_g, ω_g) are recovered from $(\hat{\vartheta}_1, \hat{\vartheta}_2)$; when the implied values are infeasible (imaginary or $\sigma_g < 1$), a grid search over admissible values minimises the GMM objective. In the model's notation the trade elasticity is $\theta_g = \sigma_g - 1$.

Alternatives (Table A.13). **Caliendo and Parro (2015)** estimate sectoral trade elasticities from tariff variation using "tetrads": ratios of trade flows across three countries in which importer, exporter and pair-symmetric unobservables cancel, so that the remaining variation in tariffs identifies θ. **Fontagné, Guimbard and Orefice (2019)** estimate HS-6 product-level elasticities from bilateral applied tariffs with fixed effects. Elasticities are noisy and possibly endogenous (they come from US data and are applied to every importer), which is why every result is also reported with *unweighted* power.

12 The engagement measure (ICEWS) (T11)

The Integrated Crisis Early Warning System (ICEWS) codes news stories automatically into "who did what to whom" events using the CAMEO event ontology; each event type carries an intensity score from −10 (most conflictual) to +10 (most cooperative), on a Goldstein-type scale

(the categories used are listed in Table A.1). The paper uses 6,733,036 bilateral events among 23,516 country pairs, 2001–2021: about 14 events per pair-year on average, with 42.96% of pairs never engaging and the top decile averaging about 145 events a year. Most events are *not* about trade: appeals for cooperation, offers to mediate, ministerial negotiations, requests to meet.

Construction of the engagement index

1. **Undirected**: the initiator is often unclear (two countries meeting reflects a joint decision), so both directions are merged.
2. **Non-violent events**: keep intensities in $[-7, 8)$, i.e. 92% of all events (violent events are studied separately).
3. **Counts and standardisation**: count events $N_{p,b,t}$ for pair p, unit-intensity bin b and year t; standardise *within pair*, $z_{p,b,t} = \left(N_{p,b,t} - \overline{N}_{p,b}\right) / \mathrm{sd}_p(N_{p,b})$, to neutralise differences in media coverage across pairs.
4. **Index**: $\text{Engagement}_{p,t} = \sum_b z_{p,b,t}$.

Two examples illustrate the raw data (Figure A.4): Japan–Thailand engagement spikes around free-trade-agreement talks (2002–03) and in August 2011, when Prime Minister Yingluck Shinawatra sought a Japanese visa for her exiled brother; Italy–Russia engagement spikes in June 2006 (Italian support for Russia's gas cut-off to Ukraine) and June 2015 (President Putin's visit to Rome to discourage support for EU sanctions). Engagement can be cooperative or adversarial; the paper therefore studies its *amount*, not its direction.

Variants (Table A.9, Panel D): standardising the total event count instead of summing bin-level z-scores (D.1), including all intensities $[-10, 10]$ (D.2), and de-duplicating events reported by several outlets at the pair × event-type × day level (D.3).

Watch out

The introduction mentions "19 million bilateral events", while Section 4.1 reports 6,733,036 events for 2001–2021 — presumably different stages of filtering. Because the index sums about fifteen bin-level z-scores, its within-pair standard deviation exceeds one; coefficients are in index units (see T10).

13 The geopolitical-alignment measure (overview)

Studying how power responds to alignment requires a measure that (i) varies both across pairs and over time, (ii) covers all pairs, and (iii) does *not* use trade or trade-policy information — which rules out UNGA voting, since many resolutions concern trade. The paper combines:

- the **Gallup World Poll**: about 1,000 respondents per country-year in 164 countries rate the leadership of the US, Russia and China. Disapproval rates are treated as noisy "distances" to three anchors, and each country is located in a two-dimensional plane by *statistical multilateration* — the logic of GPS (T22–T24);
- the **Polity** score of democratic vs. autocratic institutions, which moves slowly and anchors cross-sectional differences.

With normalised distances,

$$\text{Alignment}_{ijt} = 1 - \left[\alpha\, d_{ijt}^{\text{Gallup}} + (1-\alpha)\, d_{ijt}^{\text{Polity}}\right], \qquad \alpha = 0.5, \tag{17}$$

where d_{ijt}^{Gallup} is the Wasserstein-2 distance between the two countries' posterior location distributions.

Validation. (i) Former Soviet states' alignment with Ukraine drops sharply after the 2014 annexation of Crimea (Figure A.15), although these bilateral ties are never observed directly — only inferred from attitudes towards the three anchors. (ii) Alignment predicts UNGA vote similarity (coefficient .064 without fixed effects, .017 with pair and year effects, Table A.17); the association is mostly cross-sectional, and an ANOVA decomposition suggests the new measure contains at least six times more over-time variation than UNGA-based alignment.

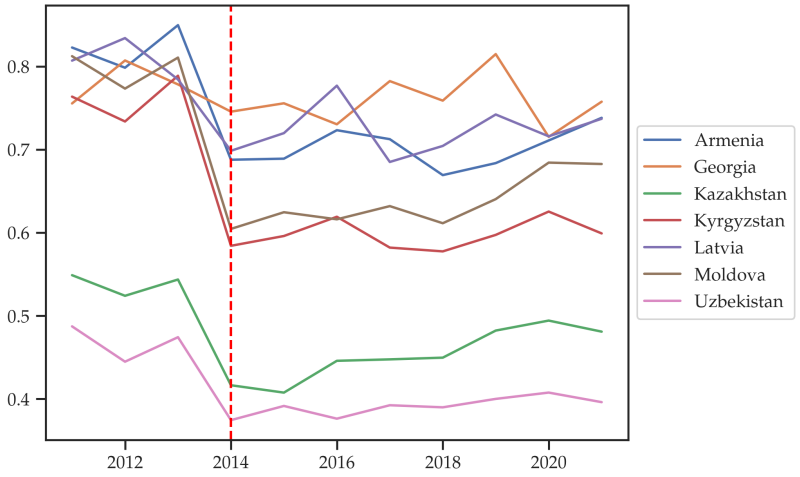


Figure 8: Former Soviet states' alignment with Ukraine (paper's Figure A.15); dashed line = 2014.

14 Other data sources

Table 4: Data sources used in the paper.

Source	Use
BACI (CEPII)	bilateral trade, HS-6, 1995–2021 → power, shift-share instrument, trade shares
SIPRI	arms trade (HS section 19); military expenditure
Broda–Weinstein; Caliendo–Parro; Fontagné et al.	trade elasticities
Eora26	domestic expenditure and services imputation
ICEWS	bilateral engagement events (2001–2021)
UNGA votes; Bailey–Strezhnev–Voeten ideal points	vote similarity (Sec. 4.6); validation of alignment
Gallup World Poll; Polity	geopolitical alignment
Marx–Pons–Rollet (2024)	elections and turnovers, 2012–2018
Global Trade Alert	targeted trade policies, 2008–2023
World Bank WDI; Cross-National Time-Series (CNTS)	GDP; balance tests (economy, openness, politics)
OECD ODA; World Bank IDS; FBIC	aid, sovereign debt, political bandwidth
Atlas / Hidalgo–Hausmann PCI; ITA critical-supply-chain list	product complexity; critical goods

Part IV

Evidence

15 Consequences: does power induce engagement?

15.1 Strategy 1: changes within country pairs

Because engagement is undirected, the regressor is undirected power $\text{Power}_{\{in\},t} = |\text{Power}_{i\to n,t}|$. The baseline specification is

$$\text{Engagement}_{\{in\},t} = \beta_1 \text{Power}_{\{in\},t-1} + \beta_2 \text{GDPdiff}_{\{in\},t-1} + \beta_3 \text{Trade}_{\{in\},t-1} + \gamma_t + \alpha_{\{in\}} + \epsilon_{\{in\},t}, \tag{12}$$

with pair fixed effects $\alpha_{\{in\}}$ (time-invariant features of a pair: distance, history, language, alliances), year effects γ_t (global shocks and trends in news coverage), the absolute GDP difference and total bilateral trade as controls, and all variables **standardised within pair**. Both directions of each pair are kept (which allows coercer and target effects in robustness checks), and standard errors are clustered by unordered pair, which makes inference equivalent to using one row per pair (T9).

Table 5: Power and engagement (paper's Table 2).

	Changes within country pairs			IV: sector-specific trade exposure		
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Power}_{\{in\},t-1}$	0.107***	0.106***	0.236***	0.454***	0.439***	0.697***
	(0.0115)	(0.0115)	(0.0163)	(0.0412)	(0.0412)	(0.0589)
$\text{GDP difference}_{\{in\},t-1}$		0.0860***	0.0923***		0.0709***	0.0858***
		(0.0184)	(0.0183)		(0.0191)	(0.0189)
$\text{Total trade}_{\{in\},t-1}$			−0.242***			−0.539***
			(0.0167)			(0.0406)
N	517,298			486,026		
Pair FE, year FE	yes					

Engagement: sum of within-pair z-scores of ICEWS event counts by unit-intensity bin in $[-7, 8)$. Power, GDP difference and total trade standardised within directed pair. IV sample: pairs with trade data in 1995–99. Standard errors clustered by undirected pair. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Reading the estimates. Column 3: a one-SD increase in a pair's power asymmetry (relative to that pair's own history) is followed next year by 0.236 more units of the engagement index — according to the paper, about +38.4% more engagement at the mean and +18.8% evaluated at the median. Two features stand out. First, adding the trade and GDP controls *raises* the coefficient from 0.106 to 0.236: total trade enters *negatively* (−0.242), so volume and asymmetry pull in opposite directions — it is asymmetric dependence, not interdependence per se, that comes with engagement. Second, the relationship is close to linear in standardised units (the binscatter in Figure 9); in raw units it is mildly concave (Figure A.7), and the left tail contains close allies with large and lopsided trade, such as the US and Canada. The relationship is not specific to great powers: excluding all pairs involving the US, China or Russia gives 0.219.

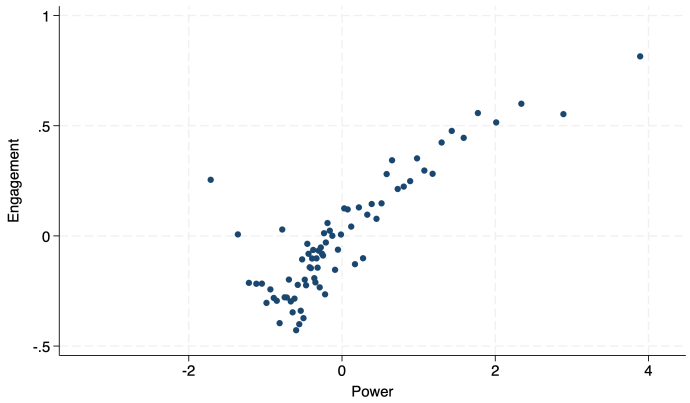


Figure 9: Binscatter of engagement on power with controls and pair and year effects (paper's Figure 2, `binsreg`; see T12).

Not sanctions, and across the spectrum. Splitting events by theme (Table A.2), power raises economic engagement unrelated to sanctions (0.0494^{***}) and non-economic engagement (0.0496^{***}), but not sanction-related events (-0.0135, n.s.). Running the regression separately for each unit-intensity bin (Table A.3), the coefficient is positive and significant in every bin from $[-8,-7)$ to $[7,8)$ (Figure 10).

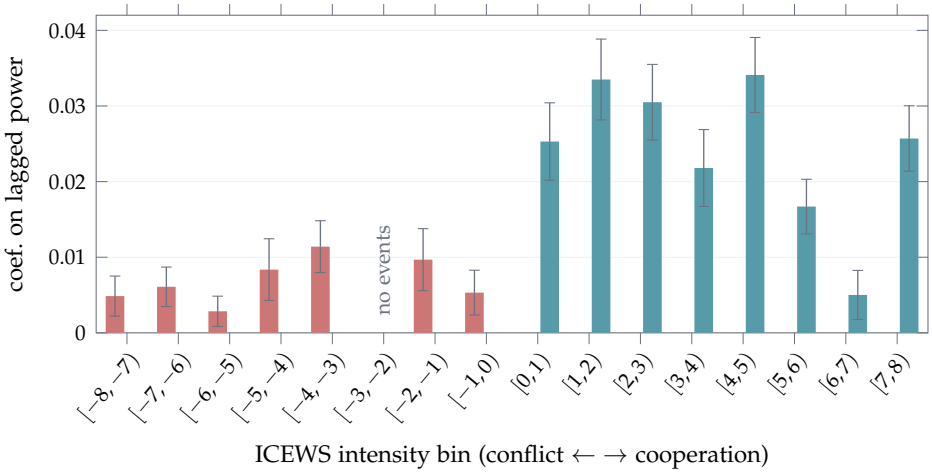


Figure 10: Coefficient on lagged power, one regression per ICEWS intensity bin (self-made from Table A.3); whiskers are 95% confidence intervals.

15.2 Strategy 2: a shift-share instrument

Within-pair changes in power could still be endogenous: countries planning to negotiate (or anticipating friction) may reshape their bilateral trade, and engagement itself may affect trade. The paper therefore isolates power changes driven by *exporters' global* export performance interacted with importers' *pre-determined* sectoral exposure.

Construction. Rewrite import dependence as a weighted sum over sectors:

$$\bar{s}_{nit} = \sum_k \underbrace{\frac{V_{nikt}}{\sum_{i'} V_{ni'kt}}}_{\hat{s}_{nikt}} \times \underbrace{\frac{\sum_{i'} V_{ni'kt}}{\sum_{i'}\sum_{k'} V_{ni'k't}}}_{w_{nkt}} \times \frac{1}{\theta_k}, \tag{13}$$

where $\hat{s}_{nikt}$ is i's share in n's imports of k and w_{nkt} is k's share in n's imports. Decompose $\hat{s}_{nikt} = \hat{s}_{ikt} + \tilde{s}_{nikt}$, where $\hat{s}_{ikt} = V_{ikt} / \sum_{i'} V_{i'kt}$ is i's *global* market share in sector k (common to all importers) and $\tilde{s}_{nikt}$ the importer-specific deviation, which may be strategic. The instrument keeps only the global component and freezes the sector weights at their 1995–1999 values:

$$b_{nit} = \sum_k \hat{s}_{ikt}\, w^{\theta}_{nk\tau}, \qquad w^{\theta}_{nk\tau} = \frac{w_{nk\tau}/\theta_k}{\sum_{k'} w_{nk'\tau}/\theta_{k'}},$$

and $|b_{nit} - b_{int}|$ instruments $\text{Power}_{\{in\},t} = |\bar{s}_{nit} - \bar{s}_{int}|$. The two-stage system is

$$\text{Power}_{\{in\},t} = \pi_1 |b_{nit} - b_{int}| + \pi_2 \text{GDPdiff}_{\{in\},t} + \pi_3 \text{Trade}_{\{in\},t} + \gamma_t + \alpha_{\{in\}} + e_{\{in\},t}, \tag{14}$$

$$\text{Engagement}_{\{in\},t} = \beta_1 \widehat{\text{Power}}_{\{in\},t-1} + \beta_2 \text{GDPdiff}_{\{in\},t-1} + \beta_3 \text{Trade}_{\{in\},t-1} + \gamma_t + \alpha_{\{in\}} + \epsilon_{\{in\},t}. \tag{15}$$

Figure 11 summarises the construction; Figure A.8 shows the variation it exploits — for instance, Germany's world market share fell by similar amounts in stone/glass/ceramics (0.513 pp) and textiles (0.436 pp), yet China's 1995–99 exposure to the two sectors differs by 11.6 percentage points.

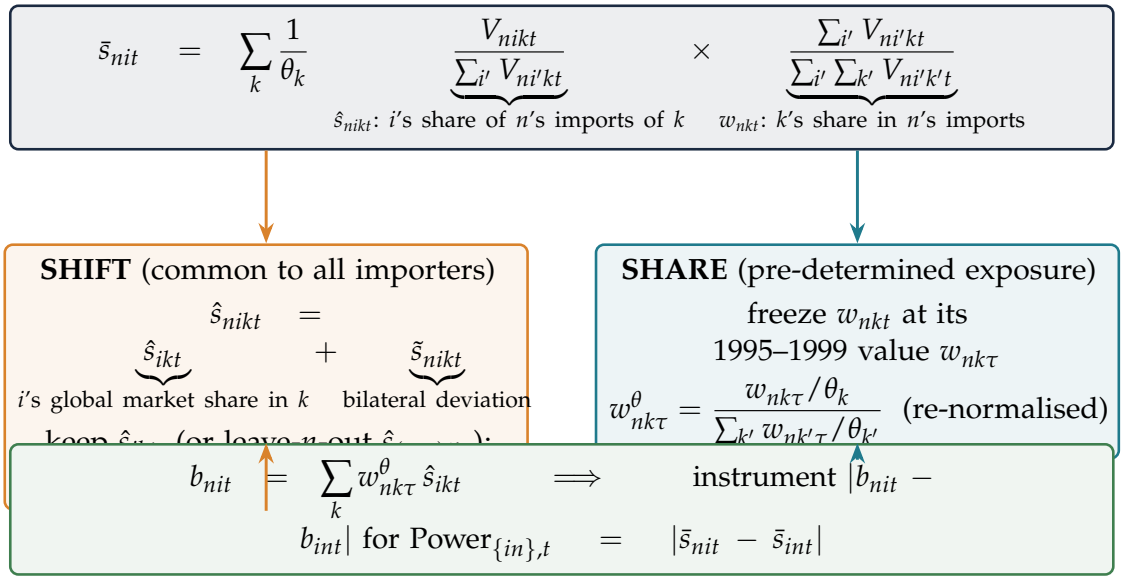


Figure 11: Building the shift-share instrument (self-made).

Implementation: the instrument (Python/pandas)

```
# pre-period sector weights w_{nk,tau}: k's share in n's imports, 1995-1999 pooled
pre = V[V.year.between(1995, 1999)].groupby(['imp', 'sec'])['value'].sum()
w = pre / pre.groupby(level='imp').transform('sum')
w_theta = w / theta.reindex(w.index.get_level_values('sec')).values
w_theta = w_theta / w_theta.groupby(level='imp').transform('sum')        # re-normalised
# shifts: exporter i's global market share in sector k and year t
X = V.groupby(['exp', 'sec', 'year'])['value'].sum()
gshare = X / X.groupby(level=['sec', 'year']).transform('sum')
# b_{nit} = sum_k w_theta[n,k] * gshare[i,k,t]
b = (w_theta.rename('w').reset_index()
       .merge(gshare.rename('g').reset_index(), on='sec')
       .assign(wg=lambda d: d.w * d.g)
       .groupby(['imp', 'exp', 'year'])['wg'].sum())
# instrument for |Power|: |b_nit - b_int| (reverse the index exactly as for power)
```

> Estimation (Stata): `ivreghdfe eng (L.abspow = L.absiv) L.gdpdiff L.trade, absorb(pair year) cluster(upair) first.`

Results. The first stage is very strong (Table A.4): coefficients 0.313, 0.314, 0.222 with F-statistics of 7,312, 7,322 and 3,996. The second stage (Table 5, columns 4–6) gives 0.454, 0.439 and 0.697, all $p < 0.01$.

Why is IV larger than OLS? Several explanations are consistent with $\hat{\beta}_{IV} \approx 3\hat{\beta}_{OLS}$: classical measurement error in power (elasticities are noisy, HS-section aggregation is coarse), which attenuates OLS; a local average treatment effect for power changes driven by exporters' global expansion, which may be more salient than idiosyncratic bilateral fluctuations; and a downward OLS bias if strategic, partner-specific trade adjustments coincide with less engagement. The paper does not adjudicate between them.

15.3 Is the shift-share instrument credible?

The paper follows the diagnostics of Goldsmith-Pinkham, Sorkin and Swift (2020) and the inference of Adão, Kolesár and Morales (2019); the methods are explained in T13–T18.

1. **Which sectors identify β?** Rotemberg weights (Table A.6, Figure 12) are concentrated: machinery and electrical equipment 0.38, textiles 0.14, plastics 0.09, miscellaneous manufactures 0.08, chemicals 0.07 (the top five sum to about 0.76); minerals has a small negative weight.
2. **Balance.** For the high-weight sectors, importers' 1995–99 shares are regressed on changes in many country characteristics (Table A.7). There are moderate associations, with mixed signs, for the top five sectors and muted ones for the others.
3. **Drop the top five sectors.** The IV estimate becomes 0.647 (0.068) (first-stage F = 2,929).
4. **Leave-one-out shifts** (Autor, Dorn and Hanson 2013): computing i's market share without its sales to n gives 0.703 (0.060).
5. **Many instruments.** Using the 21 sector instruments separately gives TSLS 480.1, MBTSLS 480.6, LIML 485.5 and HFUL 492.4, versus the Bartik estimate 431.8 (raw units, directed sample; Table A.8, Figure 13). Over-identification tests (Anderson–Rubin; Chao, Hausman, Newey, Swanson and Woutersen) reject homogeneity across instruments, but the sectors with large Rotemberg weights or strong first stages cluster around the baseline (Figure A.9); the dispersion comes from sectors that contribute little to identification.
6. **Shift-share-robust inference.** Clustering at the sector level following Adão, Kolesár and Morales gives 431.8 with standard error 12.543 (Panel C.4), versus 64.744 with pair clustering (Panel F.2).

15.4 Robustness

Figure 14 collects the standardised specifications of Table A.9. The estimate is stable when excluding the US, China and Russia (B.1), fixing the pair order (B.3), replacing pair by country effects (C.2), clustering two ways (C.3), changing the engagement definition (D.2, D.3), and using any of the alternative power measures (E.1–E.8). Three variations shrink it but keep it significant: including 1995–2000 (B.2, affected by BACI's recording change), coercer-by-year and target-by-year fixed effects (C.1, which absorb all country-specific shocks, 0.098 OLS and 0.298

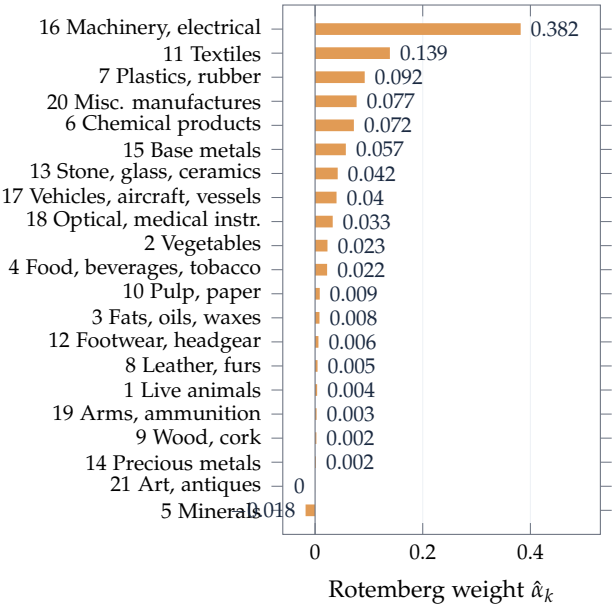


Figure 12: Rotemberg weights of the 21 HS sections (self-made from Table A.6).

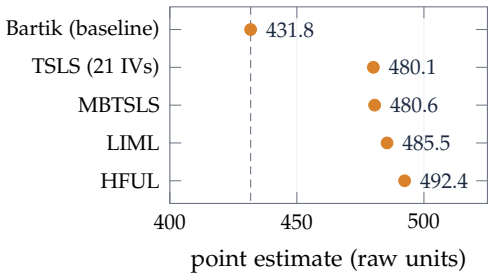


Figure 13: Bartik vs. estimators using the 21 sector instruments (self-made from Table A.8).

IV), and the level-based measure (E.6). Raw-unit versions are 105.3 (OLS) and 531.0 (IV) without standardisation (F.1), and 121.8 and 431.8 in the directed sample (F.2).

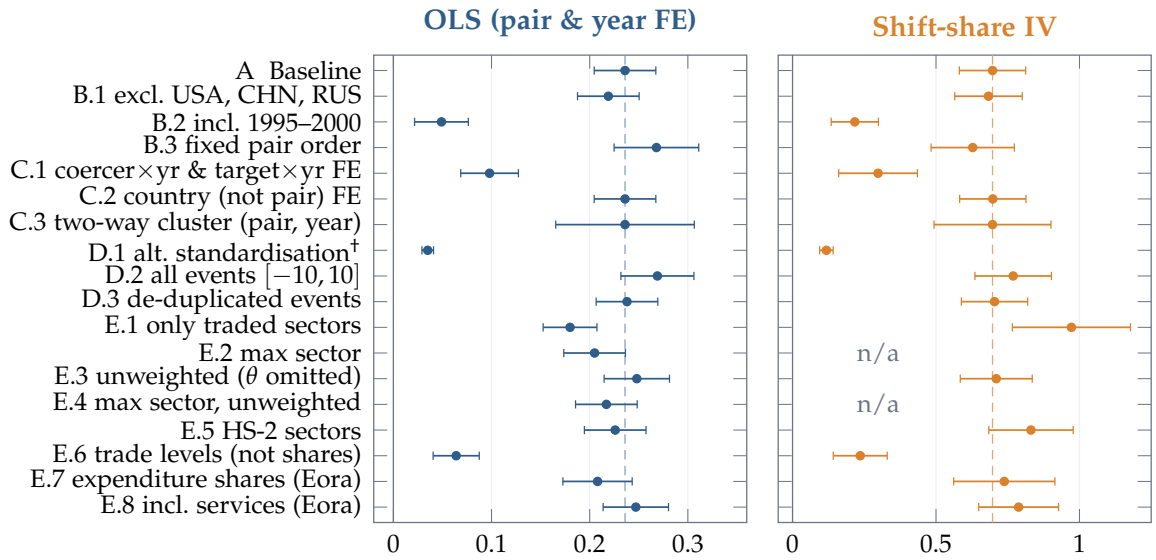


Figure 14: Robustness of the power–engagement coefficient (self-made from Table A.9); bars are 95% confidence intervals. †D.1 standardises the total event count, a different scale.

15.5 Does power change behaviour? UNGA voting

Negotiation outcomes are rarely public and the counterfactual is unobserved — as Schelling noted, whether a move is a threat or a promise depends on the baseline. UN General Assembly votes offer a partial window: all members vote on the same resolutions, so positions are comparable. Using ideal points from Bailey, Strezhnev and Voeten (2017) (T21), the paper regresses pairwise vote similarity on lagged power, controlling for lagged similarity. More power asymmetry is followed by **more similar votes**: 0.0105*** (0.00263) with OLS and 0.0350*** (0.0106) with the IV (Table A.10; lagged similarity 0.45). This is consistent with accommodation

or concessions emerging from negotiations, though not direct proof of them.

15.6 Heterogeneity: where does power bite?

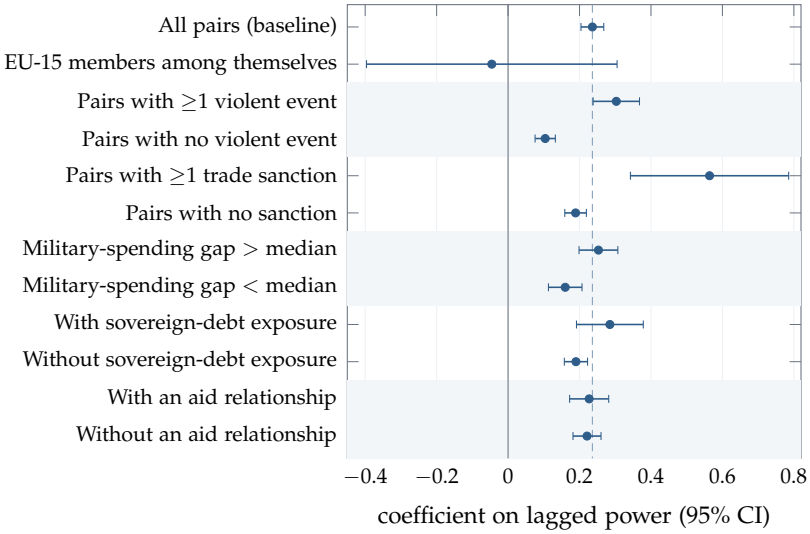


Figure 15: Heterogeneity of the power–engagement coefficient (OLS, column 3 analogues; self-made from Tables A.11, A.12 and A.16).

Institutions. Among the 15 pre-2000 EU members (105 pairs), the relationship disappears: −0.046 (s.e. 0.179) with OLS and −0.222 (0.458) with the IV (Table A.11). Re-estimating on 500 random samples of 105 EU–non-EU pairs gives a distribution centred around 0.2, with the EU-only estimate in its left tail (Figure A.10; T20). Rules on bilateral engagement inside a union can neutralise trade leverage — and the measure appears to capture *international* rather than intranational forces.

Conflict and sanctions. The effect is stronger among pairs that experienced at least one violent event (OLS 0.303 vs. 0.104; IV 0.652 vs. 0.524) or imposed at least one trade sanction (OLS 0.564 vs. 0.189; IV 1.070 vs. 0.618) (Table A.12). The paper's interpretation: under asymmetric information, realised (off-path) conflict disciplines beliefs and makes threats of trade disruption more credible.

Other sources of influence. Controlling for military-spending, aid and sovereign-debt asymmetries leaves the power coefficient essentially unchanged (0.244, 0.237, 0.227), while the effect is larger where military asymmetry is above the median (0.253 vs. 0.160) and where sovereign-debt exposure exists (0.285 vs. 0.190) (Table A.16): different sources of influence appear to be **complements**.

Sectors. Recomputing (unweighted) power one HS-4 sector at a time and re-estimating (12) gives a distribution of sector coefficients (Figure A.11) whose upper tail includes uranium ore, electrical transformers, optical fibres and semiconductors. Because these sector-specific measures omit $1/\theta_k$, the model predicts *larger* coefficients for low-elasticity sectors — which is what the data show (Figure 16; slopes -0.0022^{***} with Broda–Weinstein, -0.0013^{***} with Caliendo–Parro, -0.0018 n.s. with Fontagné et al., 18 sectors, Table A.13). Coefficients are also larger for complex products (PCI: 0.008^{***} at HS-4) and for sectors containing critical goods (0.010^{***} at HS-4;

Table A.14). At the HS-section level, chemicals, optical/medical instruments, fats and oils, pulp and paper, plastics and machinery rank highest, with largely overlapping confidence intervals (Figure A.12).

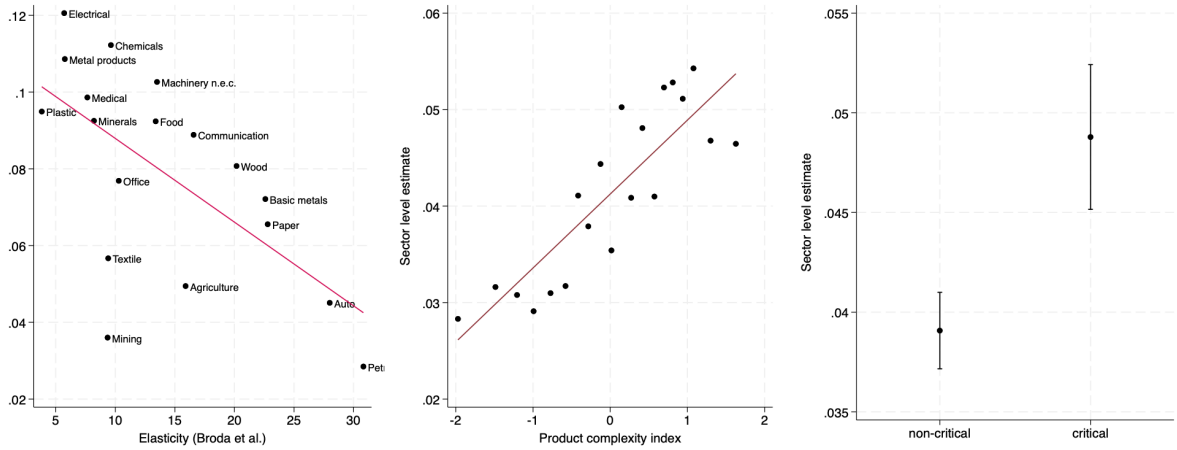


Figure 16: Sector-specific coefficients vs. trade elasticity, product complexity and critical goods (paper's Figure 3).

15.6.1 Exporter power vs. importer power

The model implies an asymmetry: a cut-off hurts the importer to first order but the exporter only to second order. Power built from the partner's *import* dependence (the baseline, "exporter power") should therefore matter more than power built from its *export* dependence ("importer power"),

$$\text{Power}^{\text{importer}}_{i\to n,t} = \sum_{k\in\mathcal{K}} \frac{1}{\theta_k} \left(\frac{V_{inkt}}{\sum_{k'} \sum_j V_{jnk't}} - \frac{V_{nikt}}{\sum_{k'} \sum_j V_{jik't}} \right). \tag{16}$$

With both in the same regression (Table A.15), exporter power is 0.234^{***} (OLS) and 0.621^{***} (IV); importer power is positive but about half as large (0.134^{***}; 0.263^{***}) and only sometimes significant in other columns. A positive importer-power coefficient can arise if threats are inframarginal or if supply chains turn one country's imports into another's exports.

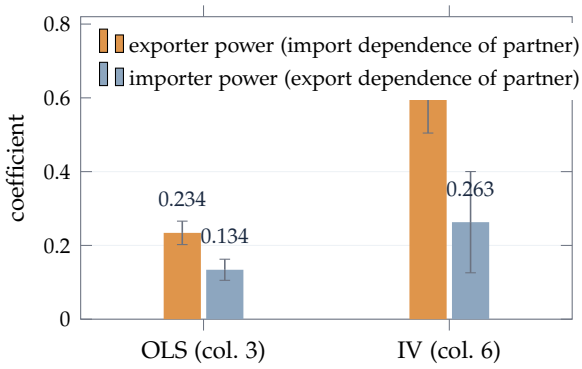


Figure 17: Exporter vs. importer power in the same regression (self-made from Table A.15).

16 Causes: do adversarial relationships prompt power build-up?

16.1 Design

Estimating how alignment affects power faces two challenges: alignment may itself respond to trade, and both countries respond to alignment shifts, so the net change in power is ambiguous. The paper uses **close elections in country A that produce a turnover** as shocks to the A–B relationship and studies B's *directed* power over A, $\text{Power}_{B\to A}$ (Figure 18).

- A's domestic electoral turnover is plausibly outside B's control and unrelated to B's politics and trade policy.
- A turnover moves A's alignment with different partners in *opposite* directions, depending on prior alignment. To respond systematically, A would need offsetting, opposite-signed bilateral policies toward many partners; B can respond to A alone with targeted policies.

Elections come from Marx, Pons and Rollet (2024): 155 presidential and 170 parliamentary elections in 177 countries, 2012–2018. An election is "close" if the winning margin is below 5%; in parliamentary systems, turnovers in which the largest party lost office through coalition formation are also included (negative margins). The resulting sample has 33 close elections, 13 of them with turnover (Table A.18). Pairs with close elections on both sides within three years are excluded (1,369 of 23,516); 14,392 pairs have no close election, 8,064 have one on one side, and 188 have elections on both sides more than three years apart.

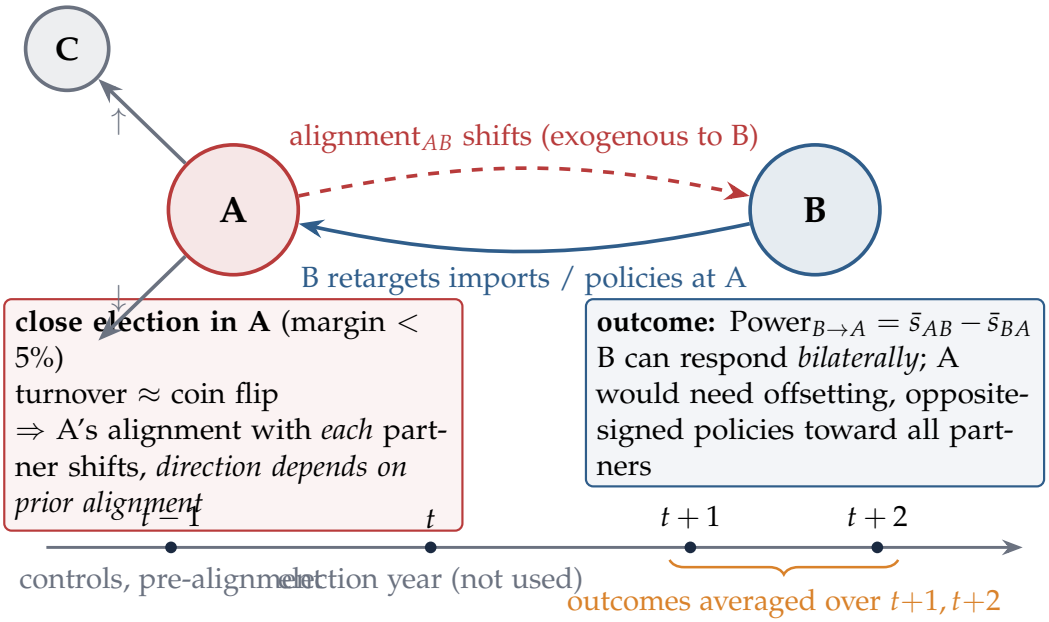


Figure 18: The identification design of Section 5 (self-made).

Two illustrations (Figure A.16): after the 2016 US election (not "close" in the electoral college, 304–227, so illustrative only), alignment with Canada and France fell while alignment with Georgia and Israel rose, with no comparable shift after the 2012 election without turnover; after Poland's close 2015 election, Canada and Germany turned away while Russia and China moved closer.

16.2 First stage

Let a_t denote the alignment of pair AB. Covariates are measured at $t-1$ and the outcome is averaged over $t+1$ and $t+2$; the election year itself is not used, because survey timing relative

to the election is unknown.

$$a_{\{t+1,t+2\}} = \beta_1 a_{t-1} + \beta_2 \text{TO}_{A,t} + \beta_3\, a_{t-1}\text{TO}_{A,t} + \beta_4 \text{NT}_{A,t} + \beta_5\, a_{t-1}\text{NT}_{A,t} + \beta_6 \text{GDPdiff}_{B\to A,t-1} + \beta_7 \text{Trade}_{t-1} + \beta_8 \text{Power}_{B\to A,t-1} + \alpha_{AB} + \gamma_t + \epsilon. \tag{18}$$

Table 6: First stage (paper's Table 3, Panel A, column 3).

Alignment_{t-1}	0.0729^{***} (0.00405)
$\text{Turnover}_{A,t}$	0.141^{***} (0.0243)
$\text{Alignment}_{t-1}\times\text{Turnover}_{A,t}$	-0.137^{***} (0.0251)
No $\text{turnover}_{A,t}$ (close election)	0.0316^{*} (0.0179)
$\text{Alignment}_{t-1}\times$No $\text{turnover}_{A,t}$	0.0205 (0.0186)
$\text{Power}_{B\to A,t-1}$	-0.00965^{**} (0.00434)
GDP $\text{difference}_{B\to A,t-1}$	-0.0203^{***} (0.00586)

Past alignment predicts future alignment when nothing happens (path dependence), and similarly after a close election without turnover; after a turnover the slope flips sign ($0.073 - 0.137 = -0.064$): **trend reversal**. Reversal does not mean deterioration: pairs that were hostile become relatively friendlier and friends drift apart (Figure 19; the raw-data version is Figure A.17).

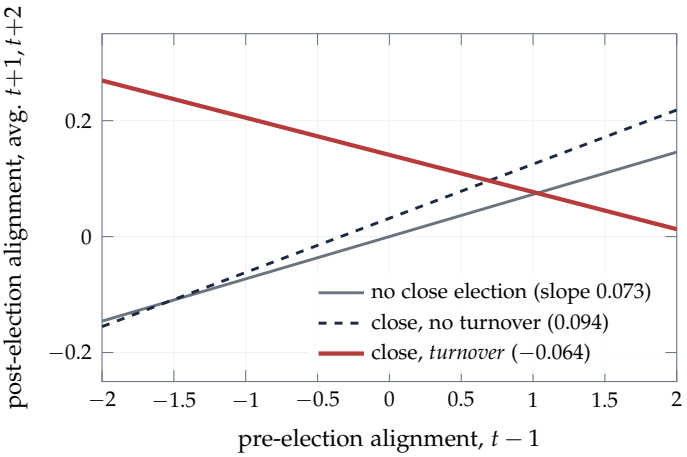


Figure 19: Implied first-stage lines from Table 3, Panel A (other covariates set to zero; self-made).

16.3 Identification checks

1. **Unanticipated, not selected.** Turnover is regressed, one variable at a time, on about 40 pairwise differences in domestic-economy, international-economy, military and political conditions at $t-1$; with false-discovery-rate adjusted q-values (Benjamini, Krieger and Yekutieli 2006; T26) none is significant (Table A.19). If an outcome were anticipated, it would already be reflected in a_{t-1}.
2. **No general policy shift by A.** Turnover predicts neither the *average* alignment (0.0187, s.e. 0.0268) nor the average power (0.0245, 0.0231) of A's partners (Table A.20): the identifying variation is the *interaction* with prior alignment.
3. **No targeted response by A.** A's targeted trade policies toward B in the Global Trade Alert do not respond (Table A.25: power-promoting -0.043, net liberal import 0.118, net liberal export 0.089, all n.s.).
4. **Arbitrary roles.** For pairs without elections the A/B labels are arbitrary; 50 random designations give coefficients close to the baseline (Figure A.18).

16.4 Main result

$$\begin{aligned}\text{Power}_{B\to A,\{t+1,t+2\}} = \beta_1 \hat{a}_{\{t+1,t+2\}} + \beta_2 a_{t-1} + \beta_3 \text{GDPdiff}_{B\to A,t-1} \\ + \beta_4 \text{Trade}_{t-1} + \beta_5 \text{Power}_{B\to A,t-1} + \alpha_{AB} + \gamma_t + \epsilon.\end{aligned} \tag{19}$$

Table 3, Panel B: second stage

	(1)	(2)	(3)
$\widehat{\text{Alignment}}_{\{t+1,t+2\}}$	-0.368^{**}	-0.381^{***}	-0.386^{***}
	(0.145)	(0.147)	(0.147)
Controls	lagged alignment, power	+ GDP gap	+ total trade

$N = 55{,}819$; pair and year effects; standard errors clustered by pair. One SD lower alignment raises B's power over A by 0.39 SD — for a pair starting at the median, a move to about the 85th percentile of power.

Countries that do not experience the turnover respond to worsening alignment by **increasing their power** over the partner: trade is at least partly strategic.

16.5 Robustness

Figure 20 summarises Table A.21. Results hold with alternative windows for alignment and power (one, two or three years after the election; the two-year/one-year combination is imprecise), with most alternative power measures, when including the 188 pairs with elections on both sides, with random A/B roles, with margin thresholds of 4, 6 and 8% (estimates grow with wider windows), with presidential elections only, and with alignment weights $\alpha = 0.25$ or 0.75. Two exceptions: power measured in *levels* gives a null (0.034), suggesting that build-up scales with size, and Eora expenditure shares give a smaller, insignificant estimate (-0.209).

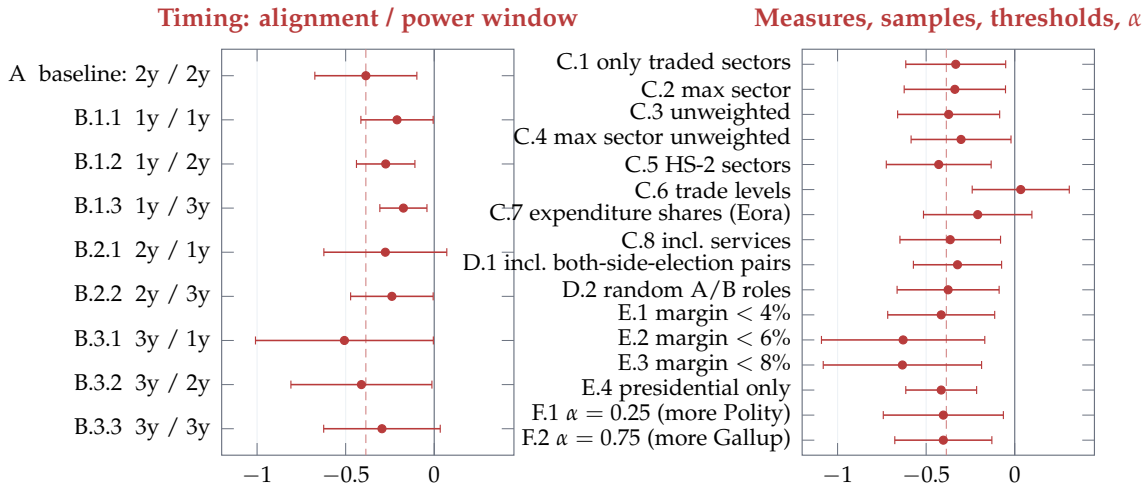


Figure 20: Robustness of the build-up result (self-made from Table A.21, column 3); bars are 95% confidence intervals.

16.6 How is power built? Trade shares and trade policies

Replacing the outcome in (19) by B's import share from A and B's export share to A (rescaled by their sample means; eq. 20), the paper finds that B's **import share** from A falls when alignment worsens (0.344^{**}, positive sign = "lower alignment, lower share"), whereas its export share barely moves (0.240, n.s.). The response is **defensive**: rather than raising exports to increase

its leverage, B reduces its dependence on A — diversifying sourcing is cheaper than quickly ramping up exports.

Using the Global Trade Alert (T28), the paper counts *targeted* policies (2,485 policies that name a target country) and defines, for flows from B to A in $t+1, t+2$: net liberal import policies (liberal minus restrictive), net liberal export policies, and *power-promoting* policies = net liberal export − net liberal import (eq. 21, controlling for the lagged outcome). Worse alignment leads B to adopt power-promoting policies (-0.380^{**}), driven by fewer liberal and more restrictive *import* measures (0.356^{***}); export policies do not move (-0.0104). A shows no targeted response (Table A.25).

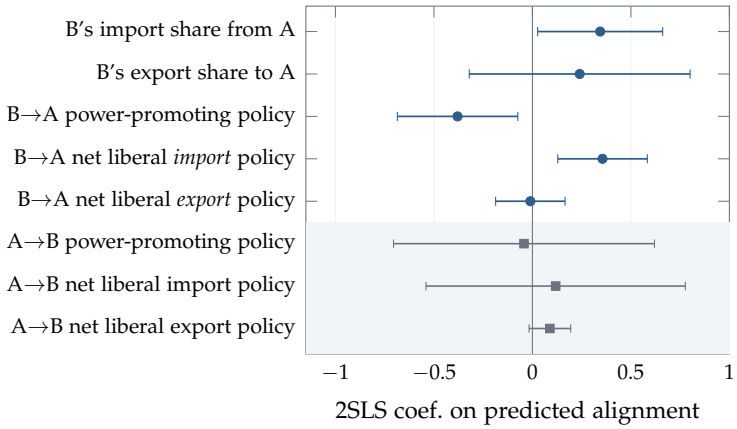


Figure 21: How power is built: 2SLS coefficients on predicted alignment for B's trade shares and policies (Table 4) and, shaded, A's own policies (Table A.25); self-made. A positive coefficient means that lower alignment lowers the outcome.

17 Why did China's power rise? (Appendix B)

B.1 Trade expansion. China's exposure measure for importer n combines China's global market share in each sector with n's 1995–99 import mix,

$$\text{ChinaExposure}_{n,t} = \sum_k \text{CGMS}_{k,t} \times w_{nk\tau}, \qquad \text{Power}_{CHN\to n,t} = \beta_1 \text{ChinaExposure}_{n,t} + X'\gamma + \epsilon. \tag{38}$$

With pair and year effects and GDP and trade controls, $\hat{\beta}_1 = 0.491^{**}$ (0.192) (Table A.22); the paper translates this into about +21.9% power for a one-SD increase in exposure, evaluated at the median. This is exactly the logic of the shift-share instrument in Section 15.2, applied to one exporter.

B.2 Industrial policy. The 10th Five-Year Plan (2001–2005) explicitly promoted sectors such as nuclear reactors, furnaces and boilers, and continued to target apparel. Regressing changes in China's exports or power at the country × HS-4 level on FYP exposure (the share of n's section imports in targeted HS-4 sectors; pre-period 2001–02, post-period 2005–06),

$$Y_{nk} = \beta_0 + \beta_1 \sum_{k\in K} \mathbf{1}[k \in \text{FYP}] \times w^K_{nk} + \alpha_n + \epsilon_{nk}, \tag{39}$$

targeted sectors saw exports rise by 25.7% and China's power by 32.1% relative to the pre-FYP level (Table A.23). The paper stresses that this does not show the plans were geopolitically motivated.

Part V

Technical toolkit

Table 7: Where each technique is explained (labels match the backup slides).

Label	Technique	Section
T1–T4	CES shares; welfare loss of a cut-off; Nash bargaining; policy wedges	4–7
T5–T7	irreversible investment; CDK general equilibrium and hat algebra; Gumbel targeted sanctions	9
T8, T11, T27	trade elasticities; ICEWS index; Eora imputations	11, 12, 10.3
T9–T10, T12	high-dimensional fixed effects, dyads, clustering; standardisation; binscatter	18–20
T13–T18	shift-share IV: identification, Rotemberg weights, balance, many-IV estimators, tests, AKM	21–26
T19–T21	sector two-step regressions, PCI, critical goods; placebo draws; UNGA ideal points	27–29
T22–T24	multilateration; rejection sampling and k-means; optimal transport	30–32
T25–T26, T28	close-election IV; false-discovery-rate control; Global Trade Alert	33, 34, 16

18 High-dimensional fixed effects, dyadic data and clustering (T9)

Why. Equation (12) has about 13,000 pair effects and 21 year effects on half a million observations; including them as dummies is wasteful, and the panel is unbalanced.

Frisch–Waugh–Lovell. The OLS coefficient on x in $y = \beta x + D\alpha + \varepsilon$ equals the coefficient from regressing $M_D y$ on $M_D x$, where M_D projects out the fixed effects. With one set of effects M_D is within-group demeaning; with two non-nested sets in an unbalanced panel there is no closed form.

Alternating projections

To compute $\ddot{v} = M_D v$ for each variable v (outcome, regressors, instrument), iterate

$$v^{(m+\frac{1}{2})} = v^{(m)} - \overline{v^{(m)}}^{\text{pair}}, \qquad v^{(m+1)} = v^{(m+\frac{1}{2})} - \overline{v^{(m+\frac{1}{2})}}^{\text{year}},$$

until the change is below a tolerance. By von Neumann's theorem on alternating projections, the iterates converge to the projection onto the orthogonal complement of the span of both sets of dummies. Then run OLS (or 2SLS) on the demeaned variables, adjusting the degrees of freedom for the absorbed effects. This is the method of Guimarães and Portugal (2010), accelerated in `reghdfe/ivreghdfe` (Correia 2016).

Dyadic data. Each unordered pair $\{i, n\}$ appears twice, as (n, i) and (i, n). Duplicating rows does not change $\hat{\beta}$ with undirected variables and, if standard errors are clustered by *unordered* pair, inference is identical to using one row per pair (paper's footnote 24); the duplication is useful because it allows coercer and target effects (Panels C.1–C.2 of Table A.9).

Two-way clustering (Cameron, Gelbach and Miller 2011) allows correlation within pairs (over time) and within years (across pairs):

$$\hat{V}_{\text{2-way}} = \hat{V}_{\text{pair}} + \hat{V}_{\text{year}} - \hat{V}_{\text{pair}\cap\text{year}}.$$

In Panel C.3 the OLS standard error rises from 0.016 to 0.036; the estimate remains significant.

Dyadic dependence

Pairs sharing a country (US–China and US–Japan) may have correlated shocks — a shock to the US affects all its dyads. Dyadic-robust standard errors (Aronow, Samii and Assenova 2015) allow correlation between any two dyads that share a member. The paper does not report them; the coercer-by-year and target-by-year effects in Panel C.1 address the same concern for the point estimate.

Implementation (Stata)

```
* one row per directed pair-year; upair = id of the unordered pair
reghdfe eng L.pow L.gdpdiff L.trade, absorb(pair year) vce(cluster upair)
reghdfe eng L.pow L.gdpdiff L.trade, absorb(pair year) vce(cluster upair year)   // two-way
reghdfe eng L.pow L.gdpdiff L.trade, absorb(coercer#year target#year) vce(cluster upair)
```

19 What a coefficient means after within-pair standardisation (T10)

Every variable is standardised within (directed) pair: $\tilde{x}_{pt} = (x_{pt} - \bar{x}_p)/\text{sd}_p(x)$.

The coefficient as an average within-pair correlation[ours]

Ignore controls and year effects. With $\tilde{x}$ and $\tilde{y}$ both standardised within pair (T_p years, sample standard deviations),

$$\hat{\beta} = \frac{\sum_p \sum_t \tilde{x}_{pt}\tilde{y}_{pt}}{\sum_p \sum_t \tilde{x}_{pt}^2} = \frac{\sum_p (T_p - 1)\,\rho_p}{\sum_p (T_p - 1)},$$

because $\sum_t \tilde{x}_{pt}^2 = T_p - 1$ and $\sum_t \tilde{x}_{pt}\tilde{y}_{pt} = (T_p - 1)\rho_p$, with ρ_p the within-pair correlation of (lagged) power and engagement. The estimate is a *length-weighted average of within-pair correlations*.

Consequences: (i) every pair counts equally, however volatile its raw power — unlike raw-unit OLS, where pairs with large swings dominate; (ii) pair effects become nearly redundant, since standardised variables already have zero mean within pair — which explains why replacing pair effects by country effects leaves the estimate at 0.236 (Panel C.2); (iii) the engagement index is a *sum* of about fifteen bin-level z-scores, so its within-pair standard deviation exceeds one and the coefficient is in index units; standardising the total event count instead gives 0.035 (Panel D.1); (iv) converting to percentages requires each pair's mean and standard deviation of raw events — the paper reports +38.4% at the mean and +18.8% at the median.

20 Binned scatter plots with controls (T12)

A binscatter visualises $\mathbb{E}[y \mid x, w]$ without assuming linearity. A common shortcut — residualise y and x on the controls, then bin the residuals — generally misrepresents the conditional mean, because residualising changes the distribution of x (Cattaneo, Crump, Farrell and Feng 2024).

Their `binsreg` instead partitions x into J quantile bins and estimates the semi-linear model

$$y_{pt} = \sum_{j=1}^{J} \mathbf{1}\{x_{pt} \in \mathcal{B}_j\}\, b_j + w'_{pt}\gamma + \alpha_p + \gamma_t + \varepsilon_{pt},$$

plotting $\hat{b}_j$ (plus the controls evaluated at a reference value) at the bin centres; J can be chosen by an integrated-MSE-optimal rule, and uniform confidence bands are available. Figure 2 of the paper (standardised variables) looks linear; Figure A.7 (raw power) shows mild concavity, as predicted by the exact loss formula in Section 5. Stata/R/Python: `binsreg eng pow gdpdiff trade, absorb(pair year)`.

21 Shift-share instruments: two ways to justify them (T13)

A shift-share ("Bartik") instrument has the form $B_\ell = \sum_k z_{\ell k} g_k$: exposure shares $z_{\ell k}$ of unit ℓ to sectors k, times sector-level shifts g_k. Here ℓ is a directed pair-year, $z_{\ell k} = w^{\theta}_{nk\tau}$ and $g_k = \hat{s}_{ikt}$ (shifts vary by exporter, sector and year).

(a) Exogenous shares (Goldsmith-Pinkham, Sorkin and Swift 2020, "GPSS"). The Bartik 2SLS estimator is numerically a GMM estimator that uses the K share-based instruments with a particular weight matrix built from the shifts. Identification then requires the shares to be (conditionally) exogenous: importers' 1995–99 import composition must be uncorrelated with later unobserved shocks to their bilateral engagement. Natural diagnostics follow: which shares matter most (Rotemberg weights), whether they correlate with observables (balance), and whether different share-instruments give similar answers (over-identification).

(b) Exogenous shocks (Borusyak, Hull and Jaravel 2022, "BHJ"). Shares may be endogenous if the shifts are as good as randomly assigned across many sectors. The 2SLS estimator can be rewritten as a weighted shock-level IV:

$$\hat{\beta} = \frac{\sum_\ell B_\ell y^{\perp}_\ell}{\sum_\ell B_\ell x^{\perp}_\ell} = \frac{\sum_k s_k g_k \bar{y}^{\perp}_k}{\sum_k s_k g_k \bar{x}^{\perp}_k}, \qquad s_k = \sum_\ell z_{\ell k}, \quad \bar{v}^{\perp}_k = \frac{\sum_\ell z_{\ell k} v^{\perp}_\ell}{s_k},$$

i.e. a regression of exposure-weighted outcomes on exposure-weighted treatments across sectors, instrumented by the shocks. Consistency requires many shocks with dispersed exposure (a small Herfindahl index $\sum_k s_k^2$); inference should be at the shock level (Section 26).

This paper argues and diagnoses in the GPSS tradition (pre-determined exposure), and reports AKM inference. Two features deserve attention:

- the instrument used in the main specification is $|b_{nit} - b_{int}|$, a non-linear function of shares and shifts; the decompositions and the AKM formula require an instrument that is linear in $z \times g$, so they are computed on the directed, raw-unit version (Panel F.2, which reproduces the estimate 431.8);
- the threat to validity is that importers with a given 1995–99 mix (e.g. heavy machinery importers) were on different geopolitical trajectories afterwards; pair effects absorb levels and year effects global trends, but not mix-specific trends — hence the balance tests.

22 Rotemberg weights (T14)

Decomposing the Bartik estimator

Let Z_k be the vector with elements $z_{\ell k} g_k$ (the sector-k component of the instrument), so $B = \sum_k Z_k$, and let $\perp$ denote residuals after partialling out controls and fixed effects. The just-identified Bartik estimator is

$$\hat{\beta}_{\text{Bartik}} = \frac{B'Y^{\perp}}{B'X^{\perp}} = \sum_k \frac{Z_k'X^{\perp}}{\sum_{k'} Z_{k'}'X^{\perp}} \cdot \frac{Z_k'Y^{\perp}}{Z_k'X^{\perp}} = \sum_k \hat{\alpha}_k \hat{\beta}_k,$$

where $\hat{\beta}_k = Z_k'Y^{\perp}/Z_k'X^{\perp}$ is the just-identified IV estimate using sector k alone and the **Rotemberg weight** $\hat{\alpha}_k = Z_k'X^{\perp}/\sum_{k'} Z_{k'}'X^{\perp}$ measures how much of the first-stage covariance sector k supplies. The weights sum to one but can be negative.

Interpretation. GPSS show that $\hat{\alpha}_k$ is the sensitivity of $\hat{\beta}_{\text{Bartik}}$ to a misspecification (a violation of exogeneity) in instrument k: the sectors with large weights are where the identifying assumption matters most, so balance tests and robustness checks should focus on them. Here the top five (machinery and electrical equipment 0.38, textiles 0.14, plastics 0.09, miscellaneous manufactures 0.08, chemicals 0.07) carry about three quarters of the weight. Figure A.9 plots each $\hat{\beta}_k$ against its first-stage F-statistic with marker size proportional to $\hat{\alpha}_k$: high-weight sectors sit close to the Bartik estimate.

Implementation (Python, after partialling out)

```python
import numpy as np
# x, y: (N,) residualised endogenous variable and outcome
# Z:    (N, K) residualised sector components z_lk * g_k (directed, raw-unit version)
a = Z.T @ x                    # first-stage covariance supplied by each sector
alpha = a / a.sum()            # Rotemberg weights (sum to one)
beta_k = (Z.T @ y) / a         # just-identified estimate for each sector
beta_bartik = (alpha * beta_k).sum()      # equals (Z.sum(1) @ y) / (Z.sum(1) @ x)
```

GPSS provide Stata and R code (`bartik_weight`). With fixed effects, residualise every column first (Section 18).

23 Balance tests and leave-one-out shifts (T15)

Balance. For each high-weight sector k and each country characteristic X (World Bank WDI for the domestic and international economy, SIPRI for the military, CNTS for political events such as demonstrations, strikes, government crises and riots, plus summary z-score indices), the paper estimates across importers

$$w_{nk\tau} = b_0 + b_1 \Delta X_n + e_{nk},$$

where ΔX_n is the change in within-country standardised X between 1995–99 and 2017–21 (Table A.7). A large b_1 would mean that exposure to sector k predicts trends that could also move diplomacy. The top five sectors show moderate associations of mixed signs; the paper therefore re-estimates without them (IV = 0.647).

Leave-one-out shifts. If n is a large buyer of i's goods, i's global market share partly reflects n's own demand, creating a mechanical link between the shift and the unit's shocks. Following

Autor, Dorn and Hanson (2013), the shift is recomputed without sales to n:

$$\hat{s}_{(-n)ikt} = \frac{V_{ikt} - V_{nikt}}{\sum_{i'} \left(V_{i'kt} - V_{ni'kt}\right)}, \qquad b'_{nit} = \sum_{k} \hat{s}_{(-n)ikt}\, w^{\theta}_{nk\tau}.$$

The estimate barely changes (0.703; first stage 0.220, F = 3,913).

24 Estimators with many instruments (T16)

Using the 21 sector components as separate instruments yields an over-identified model, which allows testing whether they agree — but 2SLS with many instruments is biased toward OLS. With controls partialled out and $P_Z = Z(Z'Z)^{-1}Z'$, $M_Z = I - P_Z$:

The k-class family

$$\hat{\beta}(\kappa) = \left[X'(I - \kappa M_Z)X\right]^{-1} X'(I - \kappa M_Z)\, y.$$

- $\kappa = 0$: OLS. $\kappa = 1$: 2SLS. Its bias relative to OLS is roughly $1/F$ for first-stage F (Bound, Jaeger and Baker 1995), so it grows with the number of weak instruments.
- **LIML**: $\kappa_{\text{LIML}} = \min_{\beta} \dfrac{(y - X\beta)'(y - X\beta)}{(y - X\beta)'M_Z(y - X\beta)}$, the smallest root of $\det(\bar{Y}'\bar{Y} - \kappa\, \bar{Y}'M_Z\bar{Y}) = 0$ with $\bar{Y} = [y, X]$. LIML is approximately median-unbiased and consistent under many-instrument asymptotics with homoskedastic errors, but has no finite moments.
- **Fuller**: $\kappa = \kappa_{\text{LIML}} - c/(n - K)$ (typically $c = 1$), which has finite moments.
- **MBTSLS** (Kolesár, Chetty, Friedman, Glaeser and Imbens 2015): $\kappa = \dfrac{1 - L/n}{1 - K/n - L/n}$, with K excluded instruments and L exogenous regressors; consistent with many instruments, even when they have direct effects orthogonal to their first-stage effects.

Jackknife estimators. Many-instrument bias comes from the own-observation terms of P_Z: $X'P_Z y$ contains $\sum_i P_{ii}X_i y_i$, whose expectation involves $\text{Cov}(X_i, \varepsilon_i) \neq 0$. Deleting them gives JIVE-type estimators. **HFUL** (Hausman, Newey, Woutersen, Chao and Swanson 2012) is a Fuller-modified jackknife LIML that is consistent with many instruments *and* heteroskedasticity:

$$\hat{\beta}_{\text{HFUL}} = \left[X'P_Z X - \sum_i P_{ii}X_i X_i' - \tilde{\alpha}X'X\right]^{-1}\left[X'P_Z y - \sum_i P_{ii}X_i y_i - \tilde{\alpha}X'y\right],$$

where $\tilde{\alpha}$ is a Fuller-type modification of the smallest eigenvalue of the analogous jackknifed variance ratio.

In the paper (Table A.8): TSLS 480.1, MBTSLS 480.6, LIML 485.5, HFUL 492.4 versus Bartik 431.8 — the estimators agree closely, which suggests that many-instrument bias is not a concern. Software: Stata `ivreg2`/`ivreghdfe` (options `liml`, `fuller()`, `kclass()`); R package `ManyIV` (Kolesár) for MBTSLS and related estimators.

25 Over-identification and weak-instrument tests (T17)

- **Sargan / Hansen** J. Regress the 2SLS residuals on all instruments: $J = nR^2 \sim \chi^2_{K-p}$ under the null that all instruments are valid and identify the *same* coefficient (Hansen's version uses the efficient GMM criterion, robust to heteroskedasticity).

- **Anderson–Rubin (1949).** To test $H_0\colon \beta = \beta_0$, regress $y - X\beta_0$ on the instruments and test that all their coefficients are zero. The test has correct size however weak the instruments. With $K > 1$ it jointly tests β_0 and the over-identifying restrictions; inverting it gives a confidence set that may be *empty* — evidence that the instruments disagree.
- **CHNSW** (Chao, Hausman, Newey, Swanson and Woutersen 2014): a jackknife version of the Sargan statistic that removes own-observation terms and is standardised to be valid with many instruments and heteroskedasticity.
- **Instrument strength.** With a single instrument and clustered errors, the relevant statistic is the (cluster-)robust or "effective" F (Montiel Olea and Pflueger 2013); rules of thumb such as $F > 10$ (Stock and Yogo 2005) are easily met here (F between 3,996 and 7,322).

Reading a rejection

Both AR and CHNSW reject homogeneity across the 21 sector instruments. Under heterogeneous effects this is expected: each $\hat{\beta}_k$ is a different weighted average of pair-level effects, so "invalid instrument" and "different estimand" are observationally similar. The GPSS-style answer is to check that the sectors that drive the estimate (large $\hat{\alpha}_k$, high F) agree with it, which they do.

26 Shift-share-robust standard errors (T18)

Why. Units with similar exposure shares load on the same unobserved sector-level shocks, so their residuals are correlated even if they are far apart geographically. Conventional or unit-clustered standard errors ignore this and can over-reject badly; Adão, Kolesár and Morales (2019, "AKM") show severe over-rejection in placebo exercises with randomly generated shifts.

The AKM variance (just-identified case)

Let $B_\ell = \sum_s w_{\ell s}\xi_s$ and let $\ddot{B}$ be the instrument after partialling out controls. Then

$$\hat{\beta} - \beta = \frac{\sum_\ell \ddot{B}_\ell \varepsilon_\ell}{\sum_\ell \ddot{B}_\ell X_\ell}.$$

AKM show that, asymptotically, $\ddot{B}_\ell \approx \sum_s w_{\ell s}\tilde{\xi}_s$, where $\tilde{\xi}_s$ is the shock net of the (share-weighted) controls. Hence the numerator is

$$\sum_\ell \ddot{B}_\ell \varepsilon_\ell \approx \sum_s \tilde{\xi}_s \underbrace{\sum_\ell w_{\ell s}\varepsilon_\ell}_{R_s},$$

a sum over *sectors*. If shocks are independent across sectors (conditional on the shares and residuals), the variance is $\sum_s \tilde{\xi}_s^2 R_s^2$ divided by the squared denominator. The feasible estimator replaces R_s by $\hat{R}_s = \sum_\ell w_{\ell s}\hat{\varepsilon}_\ell$ and $\tilde{\xi}_s$ by the coefficients from regressing $\ddot{B}$ on the share matrix W:

$$\hat{V}_{\text{AKM}} = \frac{\sum_s \hat{\tilde{\xi}}_s^2 \hat{R}_s^2}{\left(\sum_\ell \ddot{B}_\ell X_\ell\right)^2}.$$

The "AKM0" variant imposes the null hypothesis when computing residuals and has better size; confidence intervals are obtained by test inversion.

Implementation (Python sketch)

```
import numpy as np
# Bdd: (N,) residualised instrument; X: (N,) endogenous regressor;
# e: (N,) 2SLS residuals; W: (N, S) exposure shares of each observation to each shock
xi_tilde = np.linalg.lstsq(W, Bdd, rcond=None)[0]       # (S,)
R = W.T @ e                                              # (S,)
se_akm = np.sqrt(np.sum(xi_tilde**2 * R**2)) / abs(Bdd @ X)
```

R package `ShiftShareSE` (functions `reg_ss`, `ivreg_ss`) implements AKM and AKM0.

Watch out

In the paper the AKM standard error (12.5, clustering on 21 sectors) is *smaller* than the pair-clustered one (64.7) for the same point estimate. AKM corrections usually widen confidence intervals; with only 21 shock clusters, cluster-robust variance estimators can also be biased downward. AKM0 or a wild bootstrap over sectors would be natural complements.

27 Sector-specific power, product complexity and critical goods (T19)

Two-step design. (1) For each sector k — 1,241 HS-4 sectors, 21 HS sections or 18 Caliendo–Parro sectors — treat k as the entire universe of trade, compute *unweighted* power, and re-estimate (12) to obtain $\hat{\beta}_k$. (2) Regress $\hat{\beta}_k$ on sector characteristics (elasticity, product complexity, presence or share of critical goods) with robust standard errors (Tables A.13–A.14). Because $\hat{\beta}_k$ is itself estimated with varying precision, weighting by $1/\widehat{\mathrm{se}}_k^2$ (a meta-regression) is a natural refinement; at the aggregate levels only 18–21 observations are available.

Product Complexity Index (Hidalgo and Hausmann 2009)

Let X_{cp} be exports of country c in product p. Revealed comparative advantage $\mathrm{RCA}_{cp} = \frac{X_{cp}/\sum_{p'} X_{cp'}}{\sum_{c'} X_{c'p}/\sum_{c',p'} X_{c'p'}}$ defines $M_{cp} = \mathbf{1}\{\mathrm{RCA}_{cp} \geq 1\}$, diversity $k_c = \sum_p M_{cp}$ and ubiquity $k_p = \sum_c M_{cp}$. The "method of reflections" iterates averages of diversity and ubiquity; its limit is an eigenvector problem:

$$\tilde{M}_{pp'} = \sum_c \frac{M_{cp} M_{cp'}}{k_c\, k_p}, \qquad \mathrm{PCI} = \frac{Q - \overline{Q}}{\mathrm{sd}(Q)},$$

where Q is the eigenvector of $\tilde{M}$ associated with its second-largest eigenvalue (the largest corresponds to a constant vector), signed so that complex products are less ubiquitous.

Implementation: PCI (Python)

```
import numpy as np
def pci(X):                                      # X[c, p]: exports of country c in product p
    rca = (X / X.sum(1, keepdims=True)) / (X.sum(0, keepdims=True) / X.sum())
    M = (rca >= 1).astype(float)
    kc, kp = M.sum(1), M.sum(0)
    M, kc, kp = M[kc > 0][:, kp > 0], kc[kc > 0], kp[kp > 0]
    Mt = (M.T / kp[:, None]) @ (M / kc[:, None])        # Mtilde[p, p']
    vals, vecs = np.linalg.eig(Mt)
    q = np.real(vecs[:, np.argsort(-np.real(vals))[1]])   # 2nd-largest eigenvalue
    q = (q - q.mean()) / q.std()
    return q * np.sign(np.corrcoef(q, -kp)[0, 1])         # complex = less ubiquitous
```

Critical goods (Appendix E). The US International Trade Administration's Draft List of Critical Supply Chains (prepared under Executive Order 14017, February 2021) lists 2,409 products at the HTS 8- and 10-digit level in four sectors (critical minerals and materials, energy, ICT, public health) and 22 subsectors (e.g. semiconductors, large-capacity batteries, rare-earth magnets, active pharmaceutical ingredients). An HS-6 product is classified as critical if any HTS line inside it is listed (the first six HTS digits are the HS code); HS-4 sectors get an indicator, HS sections and Caliendo–Parro sectors the critical share of 2015 trade. Critical goods made up about 35% of world trade in 2015, and 14% of HS-4 sectors contain at least one.

28 Placebo by random draws (T20)

To ask whether the null effect among the 105 EU-15 pairs is unusual, the paper draws 105 random (EU, non-EU) pairs, re-estimates (12), and repeats 500 times (Figure A.10). The logic is that of randomisation inference: the share of simulated estimates at or below the EU estimate is an empirical p-value. Because random EU–non-EU pairs differ systematically from intra-EU pairs (geography, trade intensity, institutions), the comparison is best read as a descriptive benchmark rather than a formal test.

29 UNGA ideal points: a dynamic ordinal item-response model (T21)

Bailey, Strezhnev and Voeten (2017) place each country-year on one dimension, θ_{it} (support for the US-led liberal order), estimated jointly with resolution parameters. For resolution v,

$$Z_{itv} = \beta_v \theta_{it} + \varepsilon_{itv}, \quad \varepsilon \sim N(0,1); \qquad \text{vote} = \begin{cases} \text{no} & Z_{itv} < \gamma_{1v} \\ \text{abstain} & \gamma_{1v} \le Z_{itv} < \gamma_{2v} \\ \text{yes} & Z_{itv} \ge \gamma_{2v} \end{cases}$$

an ordered probit with **discrimination** β_v and cutpoints γ_v. Raw agreement scores treat all votes alike; the IRT model learns that a resolution condemning world hunger ($\beta_v \approx 0$) says nothing about blocs, while a resolution condemning democracy discriminates sharply. **Bridging**: resolutions with identical content across years are constrained to share (β_v, γ_v), which makes positions comparable over time. Estimation is Bayesian (Gibbs sampling with data augmentation, as in Albert and Chib 1993; Johnson and Albert 1999, ch. 5); absences are treated as missing at random, and posterior means are used. The paper builds pairwise vote similarity from these positions.

30 Statistical multilateration of political positions (T22)

Idea. GPS locates a receiver from noisy distances to satellites. Here the "satellites" are China, the US and Russia, and each country's Gallup disapproval rates of the three leaderships are noisy distances to them (Figure 22).

Placing the anchors

Each year put China at $(0,0)$ and the US at $(\zeta,0)$, with ζ the US disapproval of China's leadership. Russia's distance to China is α (Russian disapproval of China) and to the US β (the average of the two mutual disapproval rates). Russia's coordinates (r_1, r_2) solve

$$r_1^2 + r_2^2 = \alpha^2, \qquad (\zeta - r_1)^2 + r_2^2 = \beta^2.$$

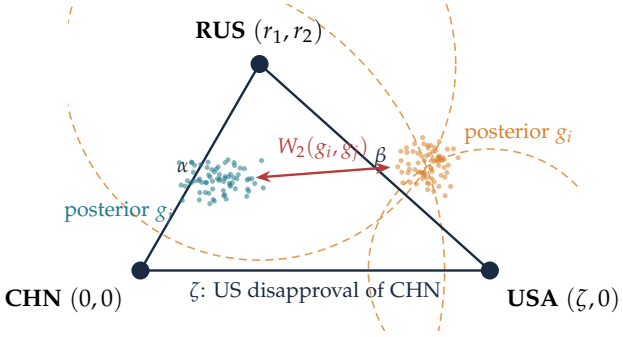


Figure 22: Statistical multilateration (self-made illustration). Dashed circles: distances reported by country i; clouds: posterior location distributions; arrow: Wasserstein distance.

Subtracting the equations gives $\zeta^2 - 2\zeta r_1 = \beta^2 - \alpha^2$, so

$$r_1 = \frac{\alpha^2 + \zeta^2 - \beta^2}{2\zeta}, \qquad r_2 = \sqrt{\alpha^2 - r_1^2} \quad \text{(positive root w.l.o.g.)}.$$

If the triangle inequality fails (2009 and 2010), the two circles do not intersect; the points closest to both circles lie on the x-axis at α and $\zeta - \beta$, and their midpoint $r_1 = (\zeta - \beta + \alpha)/2$, $r_2 = 0$ is used.

Likelihood and posterior. For any other country i in year t, reported distances are $\hat{d}_{ict} = d_{ict}(x,y) + \epsilon_{ict}$, with $d_{ict}(x,y) = \|(x,y) - c\|$ and $\epsilon_{ict} \overset{iid}{\sim} N(0, \sigma_t^2)$:

$$L_{it}(x,y) = \prod_{c \in \{\text{CHN,USA,RUS}\}} \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left\{ - \frac{(\hat{d}_{ict} - d_{ict}(x,y))^2}{2\sigma_t^2} \right\},$$

$$g_{it}(x,y) = \frac{L_{it}(x,y)}{\int_\Omega L_{it}}, \qquad \Omega = [-1.5, 1.5]^2,$$

which is the posterior under a uniform prior on Ω. Instead of a point estimate, each country is represented by its whole posterior — countries with ambiguous answers get diffuse clouds.

The noise variance — and a Neyman–Scott caveat[ours]

Given the best location (x_{it}^*, y_{it}^*), maximising the likelihood over σ^2 gives the mean squared residual; pooling a year's countries,

$$\hat{\sigma}_t^2 = \frac{1}{3N_t} \sum_i \sum_c \left(\hat{d}_{ict} - d_{ict}^*\right)^2 \qquad \text{(paper's eq. 56)}.$$

Each country contributes three distances but uses two free coordinates, leaving about one residual degree of freedom. Linearising d around the truth, $\sum_c (\hat{d}_{ict} - d_{ict}^*)^2/\sigma_t^2 \approx \chi_1^2$, so $\mathbb{E}[\hat{\sigma}_t^2] \approx \sigma_t^2/3$: the classic incidental-parameters bias of Neyman and Scott (1948). A degrees-of-freedom correction would divide by N_t rather than $3N_t$. Too small a σ_t^2 makes posteriors too concentrated and Wasserstein distances closer to distances between point estimates.

Data details. Gallup data start in 2006 but are consistent only from 2011; US respondents' answers are missing for 2008 and 2012 and are imputed with a Lasso prediction from the Pew Global Attitudes survey; Polity scores end in 2018 and are carried forward to 2021.

31 Rejection sampling and k-means compression (T23)

Rejection sampling. Draw $u \sim \text{Unif}(\Omega)$ and accept it with probability $L_{it}(u)/L_{\max}$, where $L_{\max} \geq \max L_{it}$. The density of accepted draws is proportional to $\text{Unif}(u) \cdot L_{it}(u)/L_{\max} \propto L_{it}(u)$, i.e. exactly g_{it}; the acceptance rate is $\int_\Omega L_{it}/(|\Omega| L_{\max})$, high enough in two dimensions. The paper keeps 100,000 draws per country-year.

Compression. Computing optimal-transport distances between 100,000-point clouds for every pair and year is too expensive, so each cloud is compressed with k-means (1,000 clusters): atoms are the centroids and weights the cluster shares.

Why k-means is the right compression for W_2 [ours]

Let $\hat{\mu}$ put mass $1/n$ on each draw x_m and let $\hat{\nu}$ put mass n_j/n on centroid c_j. Sending each draw to its own centroid is a feasible transport plan between $\hat{\mu}$ and $\hat{\nu}$, so

$$W_2^2(\hat{\mu}, \hat{\nu}) \leq \frac{1}{n} \sum_m \|x_m - c(x_m)\|^2,$$

which is exactly the k-means objective divided by n. Minimising the within-cluster sum of squares therefore minimises an upper bound on the compression error. Quantile binning along each axis separately ignores the two-dimensional geometry — consistent with the paper's finding that k-means gives a smaller distance to the master sample.

32 Wasserstein distance and optimal transport (T24)

For discrete distributions $\mu = \sum_a p_a \delta_{x_a}$ and $\nu = \sum_b q_b \delta_{y_b}$,

$$W_2^2(\mu, \nu) = \min_{T \geq 0} \sum_{a,b} T_{ab} \, \|x_a - y_b\|^2 \quad \text{s.t.} \quad \sum_b T_{ab} = p_a \;\; \forall a, \qquad \sum_a T_{ab} = q_b \;\; \forall b :$$

the cheapest way to move the mass of μ onto ν when moving mass costs squared distance. It is a linear program — with 1,000 × 1,000 atoms, a million variables — solved exactly by the network simplex algorithm (`ot.emd2` in the Python Optimal Transport library). Distances involving the anchors use point masses, for which W_2 is simply the root-mean-square distance between the cloud and the anchor.

Why not the Kullback–Leibler divergence? $\text{KL}(\mu \| \nu) = \sum \mu \log(\mu/\nu)$ is infinite whenever μ puts mass where ν does not — always the case for two empirical point clouds; it is asymmetric; and it ignores geometry: two non-overlapping clouds have the same (infinite) KL whether they are 0.1 or 10 apart. W_2 is a true metric and grows with how far mass has to travel. For Gaussians there is a closed form, $W_2^2 = \|m_1 - m_2\|^2 + \text{tr}\big(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}\big)$: the distance between means plus a term for differences in spread.

Implementation: the alignment pipeline (Python sketch)

```
import numpy as np, ot                          # ot = Python Optimal Transport (POT)
from sklearn.cluster import KMeans
rng = np.random.default_rng(0)

def loglik(xy, dhat, anchors, s2):              # xy: (M,2); dhat: (3,); anchors: (3,2)
    d = np.linalg.norm(xy[:, None, :] - anchors[None, :, :], axis=2)
    return -0.5 * ((dhat[None, :] - d) ** 2).sum(1) / s2
```

```
def posterior_atoms(dhat, anchors, s2, n_keep=100_000, k=1000):
    g = np.linspace(-1.5, 1.5, 601)          # fine grid to find the maximum of L
    grid = np.array(np.meshgrid(g, g)).reshape(2, -1).T
    lmax = loglik(grid, dhat, anchors, s2).max()
    kept = []
    while sum(len(d) for d in kept) < n_keep:  # rejection sampling, uniform prior
        u = rng.uniform(-1.5, 1.5, size=(200_000, 2))
        acc = np.log(rng.uniform(size=len(u))) < loglik(u, dhat, anchors, s2) - lmax
        kept.append(u[acc])
    D = np.vstack(kept)[:n_keep]
    km = KMeans(n_clusters=k, n_init=1, random_state=0).fit(D)
    w = np.bincount(km.labels_, minlength=k) / len(D)
    return km.cluster_centers_, w

def w2(atoms_i, w_i, atoms_j, w_j):
    M = ot.dist(atoms_i, atoms_j)            # squared Euclidean cost matrix
    return np.sqrt(ot.emd2(w_i, w_j, M))     # exact OT cost, then square root

# alignment_ij = 1 - [alpha * d_gallup_ij + (1 - alpha) * d_polity_ij], alpha = 0.5,
# with both distances normalised to [0, 1] before combining.
```

33 The close-election IV with an interacted first stage (T25)

Structure. The endogenous regressor is future alignment $a_{\{t+1,t+2\}}$. As written in equations (18)–(19), the four election terms — $\text{TO}_{A,t}$, $\text{TO}_{A,t} \times a_{t-1}$, $\text{NT}_{A,t}$, $\text{NT}_{A,t} \times a_{t-1}$ — appear only in the first stage, so they are the excluded instruments; lagged alignment, GDP gap, total trade, lagged power and pair and year effects are included in both stages.

Why the interaction? A turnover moves alignment in opposite directions for partners with high and low prior alignment, so its main effect averages out (Table A.20). The identifying variation is the *reversal of persistence* captured by $\text{TO} \times a_{t-1}$ — a "trend-break" design rather than a level shift.

Assumptions. (i) Within close races, which side wins is as good as random (the logic of regression-discontinuity designs on elections, Lee 2008; Marx, Pons and Rollet 2024 for turnovers worldwide); (ii) the turnover affects B's power over A only through the change in alignment (exclusion), supported by the absence of average effects and of targeted policies by A. The design is not a sharp RD — there is no running-variable polynomial; "close" means a margin below 5%, plus parliamentary turnovers with negative margins — so the robustness checks with 4%, 6% and 8% windows and presidential elections only are important.

Inference

The shocks are election-level: only 13 turnovers (among 33 close elections) generate the reversal, each affecting about 150 partners, while standard errors are clustered by pair. Clustering by the electing country A (with a wild cluster bootstrap, given few clusters) would be a natural check. Evidence on sorting in close elections is reassuring outside the US House (Eggers et al. 2015), but with few events, balance tests have limited power (see below).

Implementation (Stata)

```
* one row per (B, A, election year t); outcomes averaged over t+1, t+2; covariates at t-1
gen TOxali = TO * ali_pre
gen NTxali = NT * ali_pre
ivreghdfe powBA_post (ali_post = TO TOxali NT NTxali) ali_pre gdpdiff_pre trade_pre
    powBA_pre, ///
    absorb(pair year) cluster(pair) first
```

34 Multiple testing: false discovery rate (T26)

Table A.19 runs about 40 balance regressions; at the 5% level, two would be "significant" by chance. Controlling the family-wise error rate (Bonferroni) is very conservative; the **false discovery rate** — the expected share of false rejections among all rejections — is a more powerful criterion.

Benjamini–Hochberg and the two-stage BKY procedure

BH (1995) at level q: sort $p_{(1)} \le \cdots \le p_{(m)}$, find $k^* = \max\{k : \; p_{(k)} \le kq/m\}$ and reject $H_{(1)}, \ldots, H_{(k^*)}$.
BKY (Benjamini, Krieger and Yekutieli 2006) estimates the number of true nulls to gain power:

1. run BH at level $q' = q/(1+q)$; let r_1 be the number of rejections;
2. if $r_1 = 0$ reject nothing; if $r_1 = m$ reject everything;
3. otherwise set $\hat{m}_0 = m - r_1$ and run BH at level $q^* = q' m/\hat{m}_0$.

Sharpened q-values (Anderson 2008) report, for each hypothesis, the smallest q at which it is rejected.

Implementation (Python)

```
import numpy as np
def bh_reject(p, q):
    m = len(p); order = np.argsort(p)
    ok = p[order] <= q * np.arange(1, m + 1) / m
    k = np.max(np.nonzero(ok)[0]) + 1 if ok.any() else 0
    rej = np.zeros(m, bool); rej[order[:k]] = True
    return rej
def bky_reject(p, q):
    q1 = q / (1 + q); r1 = bh_reject(p, q1).sum(); m = len(p)
    if r1 == 0: return np.zeros(m, bool)
    if r1 == m: return np.ones(m, bool)
    return bh_reject(p, q1 * m / (m - r1))
def sharpened_q(p):
    qv = np.ones(len(p))
    for q in np.arange(1.0, 0.0, -0.001):       # from 1 down to 0.001
        qv[bky_reject(p, q)] = q                # last (smallest) q that rejects
    return qv
```

Watch out

"Nothing significant after FDR adjustment" is weak evidence of balance when the number of events is small: in Table A.19 several unadjusted t-statistics are around 2–2.6 (trade/GDP, riots, taxes on international trade, arms exports). Reporting the magnitudes, or a joint test,

complements the FDR-adjusted stars.

Part VI

Discussion

35 What the paper achieves

- **A measurable concept.** It turns Hirschman's verbal theory into a single, model-based number — elasticity-weighted asymmetric import dependence — that can be computed for every pair of countries and every year from public trade data, and shows that it is distinct from size, trade volume and imbalances.
- **Two causal questions, two research designs.** Consequences are identified with a shift-share instrument and the full modern diagnostic toolkit (Rotemberg weights, balance, leave-one-out shifts, many-instrument estimators, shift-share-robust inference); causes with a close-election design that isolates one side's response, an elegant answer to the two-sided-response problem the model itself highlights.
- **Model-driven validation.** The sector pattern (stronger effects for low-elasticity, complex and critical goods) and the exporter-vs-importer asymmetry are predictions specific to the hold-up mechanism, not generic correlations.
- **New data infrastructure**: bilateral power, a standardised engagement database, and a high-frequency alignment measure built without trade information.

36 Questions for discussion

1. **Elasticities.** The weights $1/\theta_k$ come from US import data (Broda–Weinstein) and are applied to every importer; the model's θ is $\sigma - 1$, so the mapping matters for relative weights when σ is small. How much does the cross-country heterogeneity of substitutability matter? (The unweighted variant is reassuring.)
2. **Aggregation and non-linearity.** HS sections average away choke points — rare-earth dependence lives at HS-6. The first-order formula understates losses at extreme dependence (Figure 5). Would a measure at HS-6 with the exact loss formula sharpen the results?
3. **Outcome scale.** The engagement index sums about fifteen within-pair z-scores; "standard deviation" statements and the 38%/19% conversions depend on that construction (Panel D.1 gives 0.035 when the total event count is standardised). What is the effect in events per year for a typical pair?
4. **Shift-share design.** Twenty-one sectors, machinery with 38% of the Rotemberg weight, moderate imbalance in top sectors, and over-identification tests that reject homogeneity. The AKM standard error is smaller than the pair-clustered one — unusual with only 21 shock clusters.
5. **Close-election design.** Thirteen turnovers drive the first stage; the shocks are election-level while standard errors are pair-clustered; close elections *without* turnover also enter as instruments. How do results look with standard errors clustered by electing country, or using turnover terms only?

6. **Measurement of alignment.** The noise variance in the multilateration is estimated with about one residual degree of freedom per country (Section 30); a downward-biased variance makes posteriors too tight. Does the choice affect the Wasserstein distances materially?
7. **Interpretation of engagement.** More engagement is consistent with coercion and with cooperation (Schelling's baseline problem). What share of the response is bargaining over concessions, and who concedes? UNGA convergence is a first step.
8. **Strategy and welfare.** The model's optimal policy taxes imports from adversaries. The evidence shows countries doing exactly that — defensively. Is observed de-risking close to the model's optimum, or excessive? What does the negative-sum nature of power accumulation imply for the welfare cost of geopolitical fragmentation?

37 Extensions the paper itself points to

Networks and indirect influence through third countries and supply chains; blocs and international organisations (the EU as a single actor reverses the sign of power over China); dynamics of power accumulation, retaliation and escalation; and the outcomes of negotiations, which remain unobserved.

Glossary

AKM Adão–Kolesár–Morales standard errors for shift-share designs, clustering residuals by shock (Section 26).

Anderson–Rubin test Weak-instrument-robust test of $\beta = \beta_0$; with several instruments it also tests over-identifying restrictions.

BACI CEPII's reconciled version of UN Comtrade bilateral trade at HS 6 digits.

Bartik / shift-share instrument $\sum_k (\text{exposure share})_k \times (\text{shift})_k$.

BKY Benjamini–Krieger–Yekutieli two-stage procedure controlling the false discovery rate.

CAMEO Event ontology used by ICEWS; each event type has an intensity in $[-10, 10]$.

Coercer / target The country that threatens to stop exporting / the country that depends on its imports.

CDK model Costinot–Donaldson–Komunjer multi-sector Ricardian trade model.

Exact hat algebra Computing counterfactuals in proportional changes, using only baseline shares and elasticities.

Exporter vs. importer power Power from the partner's import dependence (baseline) vs. from its export dependence.

FDR False discovery rate: expected share of false rejections among rejections.

GTA Global Trade Alert database of trade-policy interventions.

HFUL, LIML, MBTSLS Estimators robust to many instruments (Section 24).

ICEWS Integrated Crisis Early Warning System, machine-coded news events.

IRT Item-response theory model; here, ideal points from UNGA roll calls.

Leave-one-out shift Exporter's market share computed without sales to the importer in question.

Multilateration Locating a point from noisy distances to known anchors (as in GPS).

PCI Product Complexity Index (Hidalgo–Hausmann).

Power $\text{Power}_{i \to n,t} = \bar{s}_{nit} - \bar{s}_{int}$, elasticity-weighted asymmetric import dependence.

Rotemberg weight Share of the Bartik estimate attributable to one sector's instrument (Section 22).

Trade elasticity θ_k Percentage fall in relative spending for a 1% relative price increase; $\theta_k + 1$ is the elasticity of substitution.

Turnover Change of the governing party or coalition after an election.

Wasserstein distance Optimal-transport distance between distributions (Section 32).

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